

Newcomers Help Most After Being Helped, Veterans Do Not: Evidence from Stack Overflow

Abstract

Whether people who receive help online go on to help others is an open question. Prior observational studies report mixed results, and designs that ignore prior user activity may overstate reciprocity. We develop a matched difference-in-differences survival analysis of over 21 million Stack Overflow questions, comparing users whose questions were answered with propensity-score-matched users whose questions were not. Rather than asking whether reciprocity works in general, we locate the narrow conditions under which it moves behavior. Users whose questions were answered helped others slightly more often (increment at answer arrival, hazard ratio $[HR] = 1.049$, 95% CI $[1.044, 1.054]$). The response depended on tenure: newcomers responded most ($HR = 1.071$ in the first week, 95% CI $[1.056, 1.086]$), while the longest-tenured users did not respond measurably. The tenure gradient is the most robust pattern here. Timing mattered too, though far more fragilely: users answered within 30 to 60 minutes responded most strongly, while answers under 30 minutes show no elevation. Most of our data predate 2020, and the newcomer response does not survive a split at ChatGPT’s release, so we read these estimates as a pre-generative-AI baseline rather than a general test of reciprocity. Gratitude, moral obligation, social expectation, and incentive displacement remain proposed mechanisms rather than measured ones. On this one platform, generalized reciprocity works less as a general engine of cooperation than as a contributor-recruitment channel at the point of entry: designers gain most by getting newcomers’ questions answered.

Keywords: generalized reciprocity, online communities, knowledge sharing, survival analysis, platform design, Stack Overflow

1. Introduction

Whether people who receive help on an online platform go on to help others has been a persistent question in the study of online communities.

Generalized reciprocity, the tendency to help others after being helped oneself (Nowak and Roch, 2007; Baker and Bulkley, 2014), is among the mechanisms proposed to sustain cooperation on platforms like Stack Overflow and Wikipedia, where the same two people rarely trade help directly. If people who receive help reliably go on to give it, cooperation may sustain itself through cascading helping acts, even among strangers (Fowler and Christakis, 2010). The behavioral evidence, however, is inconsistent: some observational studies find substantial reciprocity effects (Baker and Bulkley, 2014; Chen et al., 2019), while others find weak or null relationships (Wasko and Faraj, 2005; Joyce and Kraut, 2006). We argue that much of this inconsistency is methodological. Our contribution is a cleaner short-run test on one platform: whether users help more after someone answers their own question. We do not ask whether generalized reciprocity sustains cooperation in general. We ask where, for whom, and under what timing it moves behavior at all, and we build the design around that narrower question. Our data do not directly measure the psychological mechanisms that theory proposes, namely gratitude, moral obligation, social expectation, and incentive displacement. What we test is the behavioral consequence those mechanisms predict.

Researchers who study reciprocity on platforms face a persistent confound: more active users are generally more likely to post questions or answers. Studies that correlate help received with help given over an aggregate period of time, without accounting for how active a user already was, conflate reciprocity with general engagement (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). Prior designs have also largely treated receiving help as a one-time event, even though people ask for and give help continuously.

We address both problems by developing a matched difference-in-differences (DiD) survival analysis, applied to over 21 million questions from Stack Overflow. For each question a user posts, we define observation windows before and after the question and compare changes in helping behavior between users whose question received an answer (treated) and propensity-score-matched users whose question did not (control). We use Cox proportional hazards models, which accommodate the continuous timing of helping and the staggered arrival of answers across users.

We develop two theoretical predictions alongside the basic reciprocity hypothesis (H1). Drawing on theories of socialization and norm internalization, we predict that the reciprocity effect is stronger among newcomers than among experienced users (H2). Newcomers have not yet internalized

community-specific norms or learned to value platform incentives such as reputation points; general moral obligations to reciprocate are therefore more likely to drive their behavior. As users accumulate experience, these general-purpose triggers may be displaced by community-specific motivations. We also predict that faster responses amplify the reciprocity effect (H3), because gratitude is more intense when help arrives while the need is still salient (McCullough et al., 2001; Tsang, 2006).

The results support the baseline reciprocity hypothesis (H1) in a modest form, and offer support for the tenure gradient (H2) and the immediacy pattern (H3). Compared with their matched controls and with their own pre-question baseline, users who receive an answer go on to answer other people’s questions more often: the pooled increment at answer arrival is $HR = 1.049$ (95% CI [1.044, 1.054], $p < .001$). How strongly users respond depends on how long they have been on the platform: newcomers respond most ($HR = 1.07$ in the first week, $p < .001$), and the longest-tenured users do not respond measurably at all. Throughout, we estimate the increment at answer arrival — the difference-in-differences contrast that removes any treated–control divergence arising before the answer. We report the waiting-period term separately, as a parallel-trends check rather than part of the effect, and we re-estimate the model under broader outcome definitions that add lower-cost contributions, namely comments and edits (Table 6).

Our study makes three contributions. First, using a design built to remove the activity confound that inflates observational estimates, we provide a cleaner short-run test: users who receive an answer help more afterward ($HR = 1.05$), but by much less than prior observational studies report. Second, reciprocity is strongest among newcomers and veterans show none. The tenure gradient is the most robust and informative pattern in the data. Third, we find an inverted U-shaped relationship in which a moderate response time carries the largest effect. Of the three, this last pattern is the most fragile, and we report it alongside the robustness check that most reduces it.

Two boundaries follow from the design, and we put them in the frame rather than in a closing caveat. The first is that the effect we identify is narrow. It is concentrated among users in their first week, it peaks for answers arriving between 30 and 60 minutes, and it is small in absolute terms even where it is strongest. The second is that the estimates are dated. Calendar year is an exact-matching stratum, so the pre-2020 period supplies most of the analysis, and the newcomer response does not survive a split at the release of ChatGPT. We therefore read our results as a pre-generative-AI

baseline for one platform rather than as a general test of whether reciprocity sustains cooperation. Reading them that way is also what makes the design worth re-running as newer data accumulate, and we discuss implications for platform designers and for future platform research in the age of generative AI.

2. Theory

2.1. Generalized Reciprocity

A central mechanism proposed to support cooperation in online communities is *generalized reciprocity*, the tendency for individuals who have received help to help others in turn, even when the recipient is not the original benefactor (Nowak and Roch, 2007; Fowler and Christakis, 2010). Unlike direct reciprocity, it requires no repeated exchange with the same person: it operates as a pay-it-forward principle, sustaining collective contributions in groups — such as online knowledge-sharing platforms — where the same two people rarely interact twice.

Socio-psychological research identifies two complementary mechanisms underlying generalized reciprocity: moral reasoning and emotional response. On the moral side, individuals reason by universalization, asking what would happen if all members requested more help than they gave. Widespread free-riding would undermine the community as a source of help, and this conclusion generates a shared obligation to reciprocate toward the group (Levine et al., 2020; Curry et al., 2019). On the emotional side, gratitude aroused by the perceived generosity of others motivates prosocial action beyond direct reciprocation (McCullough et al., 2001; Tsang, 2006; Bartlett and DeSteno, 2006) and is especially relevant where free riding is easy, as on knowledge-sharing platforms (Molm et al., 2007; Elster, 2009). These moral and emotional pathways are the *proposed* generative mechanisms behind generalized reciprocity.

A third pathway runs through visibility rather than through gratitude or obligation. Getting answered makes a user visible: the answerer is watching, others read the thread, and the asker’s profile draws traffic. Helping afterwards may then be a response to being watched rather than to being helped — a matter of *social expectation*. This pathway sits closer to indirect than to generalized reciprocity, since it works through the audience a contribution reaches rather than through the help a user received (Simpson et al., 2018; Milinski et al., 2002). We carry it alongside the other two as a proposed

mechanism, and return in Section 6.5 to what our design can and cannot say about it.

Evidence that generalized reciprocity can operate and can initiate cascading acts of generosity (Fowler and Christakis, 2010; Boyd and Richerson, 1989) is robust: comparative research indicates that the underlying norm is globally shared (Curry et al., 2019), and experimental demonstrations span laboratory (Greiner and Levati, 2005; Molm et al., 2007; Stanca, 2009) and field settings (Mujcic and Leibbrandt, 2018; Fowler and Christakis, 2010). The open question is how strongly it operates among real users on large platforms — a question observational data have answered inconsistently, for reasons we argue are largely methodological.

2.2. Empirical Challenges in Measuring Generalized Reciprocity

One challenge is that much of the evidence relies on what contributors say about their own motives, in surveys and interviews. Contributors frequently describe reciprocity as a central reason for their engagement (Wasko and Faraj, 2000; Lakhani, 2003). Yet people have difficulty articulating the moral motives behind their actions (Vaisey, 2009; McClelland et al., 1989), and introspection struggles to separate reciprocity from altruism, conformity to perceived norms, status pursuit, or the popular rhetoric of “giving back” as post-hoc rationalization (Batson, 1991; Bénabou and Tirole, 2006). Self-reported data may therefore overstate the prevalence of reciprocity as a discrete motive.

The deeper challenge is that the people who ask for help are also the people most likely to give it. Users who ask questions on platforms like Stack Overflow answer them at disproportionately high rates — not because receiving help triggered a reciprocal response, but because both behaviors reflect the same underlying characteristics, such as greater platform engagement. Studies that correlate help received with help given without accounting for this baseline activity effectively confound reciprocity with user type (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006). This confound is the central identification challenge that motivates the matched difference-in-differences design we develop in Section 4.

Prior observational studies of online reciprocity report mixed magnitudes. Baker and Bulkley (2014) and Chen et al. (2019) find positive associations between help received and subsequent helping; Wasko and Faraj (2005) and Joyce and Kraut (2006) find weak or null relationships. Our matched DiD estimates sit below the larger positive reports, consistent with the view that

designs that do not net out baseline activity overstate the effect. The tenure gradient — clear generalized reciprocity among newcomers, weaker in between, null among veterans — offers one reconciliation of the mixed prior findings.

2.3. Reciprocity and the User Trajectory

We propose that the mixed findings in the literature can be partly reconciled by attending to a dimension that prior research has not received proper attention: the user trajectory from newcomer to experienced contributor. We argue that generalized reciprocity is strongest for new users and attenuates as users gain experience on the platform.

The logic of this prediction draws on theories of socialization and norm internalization. When a user first encounters an online community, they are unfamiliar with its specific norms, social structures, and reward systems. At this stage, community-specific incentives, such as reputation points, badges, or leaderboard positions, carry little motivational weight because the newcomer has not yet learned to value them. Instead, initial contributions are guided by more general moral principles that operate independently of community-specific knowledge. Generalized reciprocity is precisely such a principle: it requires only that the individual has received help and feels a moral obligation or emotional impulse to help in return, without any need to understand or identify with the particular community (Levine et al., 2020; Curry et al., 2019).

As users interact with the community, they gradually learn its norms and develop a sense of membership. On platforms like Stack Overflow, these norms are codified in explicit reputation systems that reward helpful behavior with visible status markers. Once users internalize these community-specific norms, the broad moral obligation of reciprocity may be displaced by more localized incentive structures. The logic of universalization (“I should help because I was helped”) gives way to the logic of community membership (“I should help because that is what valued members do”). This norm displacement echoes accounts of how external incentives can crowd out moral motives (Bénabou and Tirole, 2006).

Two further mechanisms accelerate this decline. As users accumulate experiences of receiving help, each additional instance becomes less surprising, and the gratitude that fuels reciprocity fades with repetition (McCullough et al., 2008). And as users learn that the platform rewards helping with reputation and status, they may increasingly attribute others’ helpfulness to

status pursuit rather than genuine generosity, further dampening that gratitude.

This account predicts that generalized reciprocity is strongest among newcomers and declines with experience. It also helps explain why prior studies have reached different conclusions about the importance of reciprocity: those that focus on or oversample experienced users may underestimate reciprocity, while those that capture newcomer behavior may find stronger effects.

HYPOTHESIS 1 (H1). *Receiving help increases a user’s propensity to help others (generalized reciprocity effect).*

HYPOTHESIS 2 (H2). *The generalized reciprocity effect is strongest among newcomers and decreases as users gain experience on the platform.*

The stakes of this prediction extend beyond reconciling mixed findings. If the reciprocity association is strongest among newcomers, generalized reciprocity would function less as a general engine of cooperation among established members than as a *contributor-recruitment* channel operating at the point of entry, before community-specific incentives take hold — a design implication we develop in Section 6.4.

2.4. Response Time and the Immediacy of Reciprocity

The emotional mechanisms underlying reciprocity suggest that the speed of help matters for its strength. Gratitude, the primary proposed emotional driver of generalized reciprocity (McCullough et al., 2001), is triggered by the perception of a benefactor’s generosity, and its intensity is likely to depend on the characteristics of the helping experience — including its timeliness.

A swift response can amplify the experience of receiving help: a fast answer signals that the community is attentive and generous, and it arrives while the asker’s need is still salient, when gratitude should be most intense. Conversely, when an answer arrives hours or days later, the original need may have faded or been met elsewhere, the associative link between asking and receiving weakens, and the delay itself may signal an unresponsive community. Each of these changes attenuates the emotional and moral responses that underpin reciprocation.

These considerations lead to a straightforward prediction:

HYPOTHESIS 3 (H3). *Timelier responses to a user’s question strengthen the generalized reciprocity effect.*

This hypothesis follows from the dominant theoretical accounts of generalized reciprocity, which emphasize emotional immediacy and the salience of the helping experience. Those accounts imply that the advantage grows monotonically as help arrives sooner. Field settings, however, introduce session dynamics — whether the asker is still online when the answer arrives — that laboratory theory abstracts away. We therefore treat the prediction as directional, test whether it holds monotonically, and return to this qualification when presenting the results.

3. Empirical Setting: Stack Overflow

Stack Overflow is well suited to examining generalized reciprocity in online knowledge sharing. The platform publishes quarterly data dumps that record what every user did and when, reaching back to its inception. From these records we can reconstruct how people helped one another over time, and in what order they received and gave help — the sequence on which any study of generalized reciprocity depends. Stack Overflow’s active question-and-answer volume has declined sharply since the release of generative AI tools in late 2022 (del Rio-Chanona et al., 2024; Burtch et al., 2024); our sample spans platform inception through 1 April 2025 and therefore covers a period that mostly predates, but extends into, the steepest LLM-driven decline. That timing makes the sample a clean pre-substitution baseline: it records what a human answering ecosystem contributed to recruiting new contributors while that ecosystem was still intact, a measurement that may not be possible to make again. We treat the findings as a historical snapshot of one large, technical, relatively anonymous, status-driven platform; external validity to other communities is discussed in Section 6.

At its core, Stack Overflow is a question-and-answer platform for software developers and technologists. As of June 2025, shortly after our sample cut-off, the site had accumulated over 24 million questions and 36 million answers from more than 29 million registered users (Stack Exchange, 2025). Its user base spans a wide range of experience levels, from learners to professional software developers (Stack Overflow, 2024). Questions on the platform must be tightly focused on specific programming problems; broad or opinion-based inquiries are typically closed by community moderation (Stack Exchange Inc.,

2024). The most frequently discussed topics, as reflected in question tags, include JavaScript, Python, Java, C#, PHP, Android, and HTML, spanning areas from web development and mobile applications to data science and machine learning. Stack Overflow is built around several core features: tags categorize questions by topic, facilitating expertise matching; a voting system allows users to upvote or downvote questions and answers to assess quality; and reputation points are earned for valued contributions such as providing high-quality answers (Stack Exchange Inc., 2024).

For our study, the key feature is the question-and-answer interaction between users. When a user posts a question, other users may provide answers. This creates a timestamped sequence (question posted, answer received) that we exploit to measure the temporal dynamics of generalized reciprocity. On Stack Overflow, question-askers and answerers are typically different individuals, which makes the platform well suited to studying *generalized* reciprocity: users who receive help from one community member may subsequently help entirely different members.

4. Data and Methods

To test our predictions, we develop a matched staggered difference-in-differences survival analysis that exploits the order in which users ask for and give help on Stack Overflow. Our design compares changes in helping behavior (measured as the hazard of posting an answer) before and after a user receives an answer, between users who receive an answer and matched users who do not. This approach integrates two methodological traditions: the staggered difference-in-differences framework of Callaway and Sant’Anna (2021), which accommodates variation in treatment timing and treatment effect heterogeneity, and the counting-process formulation of the Cox proportional hazards model (Andersen and Gill, 1982), which allows treatment status and other covariates to vary continuously over the observation window.

4.1. Study Design

For each question posted by a user, we construct a timeline centered on two key events: the posting of the question (T_{question}) and, for questions that receive an answer, the time at which the first answer with a non-negative vote score is posted (T_{answer}). The observation window opens two days before the question ($T_{\text{question}} - 2D$) and closes two days *after the answer arrives* ($T_{\text{answer}} + 2D$). The window is therefore anchored on the answer rather than

on the question: the pre-question and post-answer segments each have a fixed two-day duration, while the intervening waiting period runs for the question’s response time. The total timeline spans $4D + RT$ (response time), so slower-answered questions have proportionally longer timelines; this ensures every treated question — however late its answer — receives a full two-day post-answer phase in which we can observe the reciprocity effect.

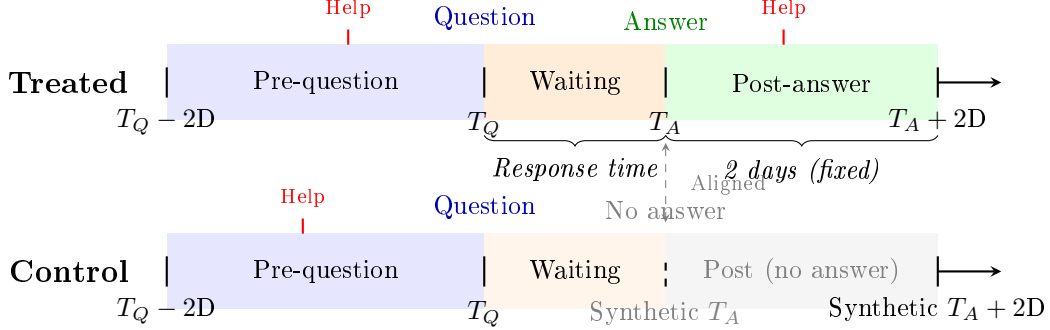
We define three phases within each observation window (see Figure 1). The first phase, the *pre-question phase*, is from the start of the observation window to the posting of the question ($T_{\text{question}} - 2D$ to T_{question}). This phase captures baseline helping activity before the focal question is asked. The second phase, the *waiting period*, is from the posting of the question to the receipt of the first answer (T_{question} to T_{answer}). During this interval, the user has asked for help but has not yet received it. The final *post-answer phase* is from the receipt of the answer to the end of the observation window (T_{answer} to $T_{\text{answer}} + 2D$), a fixed two-day span regardless of how long the answer took to arrive. This is the phase in which the reciprocity effect, if it exists, should manifest.

A key feature of our design is that the post-answer (treated) phase begins at a different elapsed time for each observation, depending on how quickly the question receives a response.

This staggered timing of the post-answer phase corresponds to the setting studied by Callaway and Sant’Anna (2021), who show that standard two-way fixed-effects estimators can be biased when treatment timing varies, and treatment effects are heterogeneous. Our approach avoids this pitfall by explicitly modeling the time-varying change in treatment status within each observation window and by comparing treated observations only to not-yet-treated or never-treated controls.

Control users (those whose questions did not receive an answer) have no natural transition between the waiting period and the post-answer phase. To maintain temporal alignment with the matched treated observation, we assign the control observation a synthetic phase transition at the same elapsed time as the treated user’s response time. That is, if the treated user’s answer was posted 4 hours after their question, the matched control observation’s timeline is split at $T_{\text{question}} + 4h$ and its window closes at $T_{\text{question}} + 4h + 2D$, creating an aligned “post” phase of the same two-day duration. Because the control inherits the treated response time, matched treated and control observations share identical window boundaries and identical two-day post-answer durations, so any differences between them in the post-answer phase

Figure 1: Study Design: Matched Difference-in-Differences Survival Analysis



Note. The observation window opens two days before the question (T_Q) and closes two days after the answer ($T_A + 2D$); it is anchored on the answer, so the pre-question and post-answer segments each last a fixed two days while the intervening waiting period spans the response time. The total window length is thus $4D + RT$ and is not drawn to scale here. The treated timeline (top) shows three phases: the pre-question phase (baseline), the waiting period (question posted, no answer yet), and the post-answer phase (after the first qualifying answer T_A arrives). The control timeline (bottom) is from a propensity-score-matched user whose question did not receive an answer. To ensure temporal alignment, the control's post-question period is split at a synthetic transition point matching the treated user's response time. Red markers indicate help events (answers posted by the focal user to other users' questions). The treatment effect is the differential change in helping hazard between treated and control observations in the post-answer phase, relative to the pre-question baseline. The staggered timing of T_A across observations motivates our use of matched DiD.

are not driven by differences in the length of the observation period.

We center the survival clock in our Cox models on the posting of the question: time zero corresponds to the start of the observation window ($T_{\text{question}} - 2D$), so the question itself lands at day 2 of the timeline. Because the window closes two days after the answer, its end falls at day $4 + RT$ rather than at a fixed day 4; a question answered within the first day therefore closes near day 4, while a question answered after three days closes near day 7. We record every help event (an answer posted by the focal user) at its exact timestamp and convert it to elapsed time on this clock.

Within each timeline, we track all instances in which the user provides an answer to another user's question. These helping events constitute the outcome in our survival analysis; the decomposition (Table 6) also examines broader definitions that count comments and edits. Conceptually, our design asks: Does the hazard of helping increase after a user receives an answer,

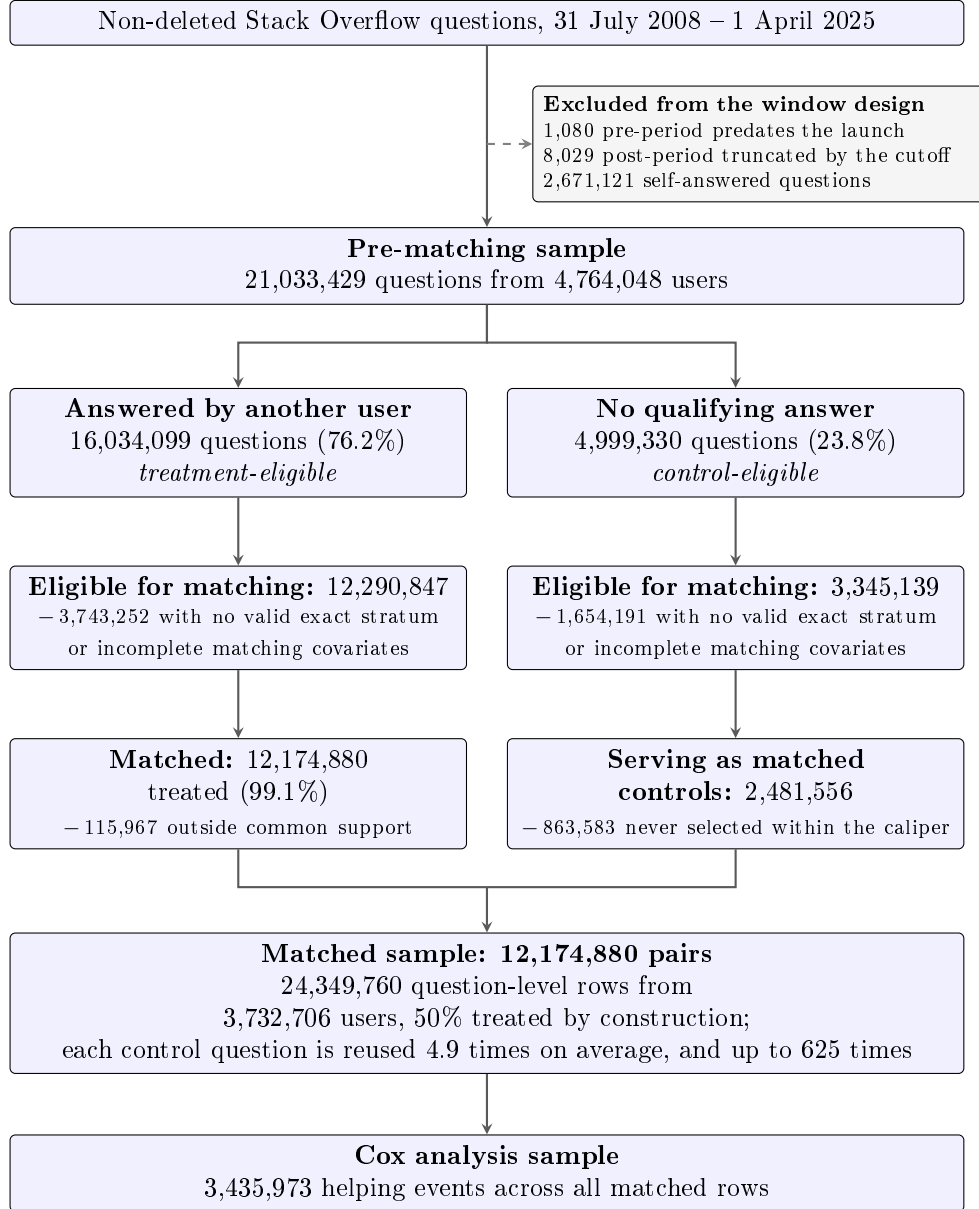
relative to the change in hazard for a matched user who does not receive an answer? We present descriptive evidence on the raw help rates over this timeline in Section 5 (Figure 3), before turning to the formal Cox model estimates.

4.2. Data

We used the full sample of non-deleted questions from Stack Overflow’s inception on 31 July 2008 through 1 April 2025. We addressed two edge cases that would render our fixed-length observation window invalid. First, for early platform questions, the pre-period could predate the platform’s launch, meaning no answers were possible at that time. We removed 1,080 such cases. Second, due to the data dump’s 1 April 2025 cutoff, the post-period could be prematurely truncated. We excluded 8,029 cases affected by this. We then excluded 2,671,121 questions that received an answer from the question poster. These filtering steps resulted in a pre-matching dataset of 21,033,429 questions from 4,764,048 unique users, of which 16,034,099 (76.2%) received at least one answer from another user (treatment-eligible) and 4,999,330 (23.8%) received no answer (control-eligible). Matching (described below) is nearest-neighbor with replacement within exact strata and requires complete matching covariates, so 12,290,847 answered and 3,345,139 unanswered questions (15,635,986 in total) were eligible. Of the 12,290,847 eligible treated questions, 12,174,880 (99.1%) were matched to a control; the remaining 115,967 fell outside common support, and a further 3,743,252 answered questions lacked a valid matching stratum or complete covariates. This attrition is not random: the treated propensity-score density piles up near one (Figure B.6), so the discarded questions are disproportionately those that were almost certain to be answered. The 12,174,880 matched treated questions draw on 2,481,556 distinct control questions, each reused about 4.9 times on average (up to 625 times) to fill the control slots. The matched file therefore comprises 24,349,760 question-level rows (12,174,880 matched pairs) from 3,732,706 unique users, with a 50% treatment rate by construction. All matched rows enter the Cox analysis; the helping outcome is the posting of an answer to another user’s question (answers to one’s own question do not count), yielding 3,435,973 helping events in the pooled analysis sample (Table 5).

Figure 2 traces these steps from the public dump to the matched analysis sample.

Figure 2: Sample Construction: From the Public Dump to the Matched Analysis Sample



Note. Counts follow the filtering described in this section, and grey text records what each step removes. Matching is nearest-neighbor with replacement within exact strata of calendar year, primary tag, and number of tags, which is why 12,174,880 matched treated questions draw on only 2,481,556 distinct control questions. Appendix C discusses what the off-support discarding implies for the scope of the estimand; Appendix E reconciles the sample sizes reported across the Cox tables.

4.3. Variables

Table 1 provides an overview of the variables used in our analysis. The phase terminology is consistent with Figure 1: the pre-question phase captures baseline behavior, the waiting period spans the interval from question to answer, and the post-answer phase captures the period after the answer arrives. We use “post-question” to refer to the combined waiting period and post-answer phase whenever distinguishing between these two sub-periods is not necessary.

Outcome. The event of interest is posting an answer to another user’s question. Within each window we track when (and whether) the focal user answers any other user’s question. The Cox model then estimates the instantaneous hazard of this helping event as a function of time-varying covariates that capture the phase of the observation timeline and the treatment status. Answering one’s own question, and self-directed acts such as accepting an answer, do not count as helping events. To address the concern that answers alone may understate lower-cost reciprocity, we re-estimate the pooled model under broader outcome definitions (comments only, edits only, and summing all these channels together) and report the full decomposition in Table 6.

We treat the posting of an answer as the theoretically central helping act. Generalized reciprocity, as modeled in the experimental literature, is a *costly* prosocial contribution: individuals pass a benefit to an unrelated third party at real expense to themselves (Greiner and Levati, 2005; Stanca, 2009; Mujic and Leibbrandt, 2018). It is the willingness to bear that cost — not mere approval or endorsement — that distinguishes reciprocity from low-cost signaling (Molm et al., 2007). Answering a stranger’s question demands substantive effort and yields a public good for the community. Lower-cost gestures such as comments, upvotes, and edits show engagement, curation, or approval but do not incur the same time commitment our theoretical construct suggests. We therefore anchor the outcome on answers and report comment- and edit-inclusive alternatives in the decomposition (Table 6); upvotes are the one lower-cost channel no Stack Overflow data can supply, because the public dump withholds voter identity. As the decomposition shows, this choice appears reasonable: comments alone carry only a slight elevation, well below the answers increment, so reciprocity manifests the strongest in answers and broadening the outcome decreases the estimate (see Section 6.5).

Treatment. The treatment is the receipt of an answer. We measure it with *treatment*, a binary variable set to one when the user’s question received,

Table 1: Overview of Variables

Variable	Description
<i>Outcome (Survival Event)</i>	
helpEvent	A binary event indicator equal to 1 at times when the user posts an answer to another user’s question within the observation window (answers to one’s own question do not count). Broader definitions adding comments and edits are reported in the outcome decomposition.
<i>Time-Varying Covariates</i>	
PostQuestion	Binary; 1 for intervals occurring after the user posts their question ($t \geq T_{\text{question}}$), 0 otherwise. Captures the combined post-question period (waiting period + post-answer phase).
PostAnswer	Binary; 1 for intervals occurring after the user receives their first answer ($t \geq T_{\text{answer}}$), 0 otherwise. For control users (no answer received), this variable switches on at the synthetic transition point at the matched treated user’s response time. Isolates the post-answer phase.
treatedPostQuestion	Interaction of treatment $\times$ phasePostQuestion; captures differential behavior of treated users in the entire period after the question.
isTreatedActive	Interaction of treatment $\times$ phasePostAnswer; the primary treatment effect capturing the additional lift in helping hazard after receiving an answer, beyond any change attributable to the act of asking a question.
ResponseTime	Time between question and the first answer (excluding down-voted answers) of the treated question. We include interactions with all four variables above.

within seven days of posting, at least one answer from *another* user carrying a non-negative score; the seven-day cap keeps the fixed two-day post-answer window inside the observable record. Control questions are the complement: those that received no such qualifying answer. This definition pools three cases that share the absence of timely, positively scored help from another user — questions never answered, questions answered within seven days only by downvoted posts, and questions whose first qualifying answer arrived after seven days. We choose receiving an answer as the treatment rather than on Stack Overflow’s “accepted answer” mechanism because accepting an answer requires the user to return to the site, and users who return are plausibly also the users who go on to help.

Response Time. For each treated observation, the response-time variable used in the interaction terms is $\log(1 + \text{RT})$, where $\text{RT} = t_{\text{answer}} - t_{\text{question}}$ is the actual response time in hours: the elapsed time from the posting of the question to the arrival of the first answer with a non-negative vote score. The origin is the question itself, so RT is the true question-to-answer latency and is unaffected by the length of the pre-question window. The discrete response-time bins reported below (Table 9) are likewise cut on this same actual response time, so a bin labelled “1–3 days” contains questions answered 24–72 hours after posting. For numerical stability, both response-time interaction terms ($\text{isTreatedActive} \times \log(1 + \text{RT})$ and $\text{treatedPostQuestion} \times \log(1 + \text{RT})$) are standardized (z-scored) and then clipped to their 5th–95th percentile range before model fitting. Consequently, the reported interaction coefficients γ and δ are expressed per one standard deviation of the respective interaction term rather than per unit of $\log(1 + \text{RT})$.

User Experience. To test Hypothesis 2, we stratify our analysis by user tenure, defined as the number of days between a user’s first activity on the platform and the start of the observation window. We define seven tenure buckets using right-closed intervals: $[0, 7]$, $(7, 30]$, $(30, 180]$, $(180, 365]$, $(365, 1095]$, $(1095, 2190]$, and $(2190, \infty)$ days, corresponding to the labels less than one week, one week to one month, one to six months, six to twelve months, one to three years, three to six years, and more than six years, respectively. Running separate models by bucket lets us examine how the reciprocity effect varies across the user trajectory without imposing a parametric functional form.

4.4. Propensity Score Matching

Receiving an answer (the treatment) is correlated with user characteristics: more experienced users, for instance, might ask higher-quality questions that are more likely to be answered. We therefore employ propensity score matching (Rosenbaum and Rubin, 1983) to improve the comparability of the treatment and control groups. Importantly, we match *across different users*: we pair each treated question with a control question from a *different* user, chosen for its similar propensity to receive an answer based on observable pre-treatment characteristics. This cross-user matching ensures that treated and control observations are independent conditional on the matched covariates, while the DiD structure within each pair absorbs residual differences.

We estimate the propensity score using pre-treatment covariates that jointly capture user experience, recent activity, and question characteristics, as listed in Table 2. The covariate *tagAcceptShareAvg* addresses the concern that question complexity may confound the treatment: questions in tags with low accepted-answer rates are inherently more difficult, and matching on this rate ensures that treated and control questions come from similarly tractable topic areas. Matching on the tag-level accepted-answer rate also matters for the response-time analysis, because it mitigates the concern that slower response times simply proxy for differently sized sub-communities with lower throughput. We use nearest-neighbor matching *with replacement* and a caliper of 0.05 on the propensity score, requiring in addition an exact match on the calendar year, the question’s primary tag, and the number of tags. Because the pool of unanswered questions is smaller than the pool of answered questions, some control questions are matched to more than one treated question; the matched file therefore contains more rows than unique question pairs (12.2 million pairs, 24.3 million rows). Each row enters the Cox analysis as an independent observation window; standard errors are discussed in the robustness appendix.

Table 3 displays the covariate balance before and after matching. Before matching, several covariates showed a meaningful imbalance between the treated and control groups. For instance, treated users had asked more questions in the prior 30 days ($SMD = 0.197$) and 7 days ($SMD = 0.174$), reflecting that users who ask questions that attract answers tend to be more recently active. They also posted longer questions and had higher reputation (body length $SMD = -0.183$; owner reputation $SMD = 0.115$). The largest imbalance was in the tag-level accepted-answer rate ($SMD = 0.695$), confirming that question topic characteristics strongly predict answer receipt

Table 2: Definition of Matching Covariates

Variable name	Definition
<i>Exact matching</i>	
calendarYear	Calendar year exact matching indicating the year of the observation, controlling for secular trends in platform activity and answer rates.
topLevelTag	First tag of the question, indicating the main topic and the community that sees the question.
numTags	Number of tags on the question. Treated and control questions are matched on the exact tag count, so they pose comparably scoped problems.
<i>Propensity score matching</i>	
userTenure	Number of days since the user’s first recorded activity on the platform, capturing cumulative experience.
numQuestionsAskedAT	Total number of questions asked by the user before the observation window.
numHelpProvidedAT	Total number of answers provided by the user before the observation window.
numQuestionsAsked30D	Number of questions asked in the 30 days preceding the observation window.
numHelpProvided30D	Number of answers provided in the 30 days preceding the observation window.
numQuestionsAsked7D	Number of questions asked in the 7 days preceding the observation window.
numHelpProvided7D	Number of answers provided in the 7 days preceding the observation window.
tagAcceptShareAvg	Average share of questions that receive an <i>accepted</i> answer within the question’s tags, capturing topic-specific difficulty.
<i>Observable question and author characteristics (propensity score matching)</i>	
postHour	Hour of day at which the question was posted, capturing diurnal variation in who is online to answer.
postDayOfWeek	Day of week on which the question was posted.
bodyLenChars	Length of the question body in characters, a proxy for question detail and effort.
titleLenChars	Length of the question title in characters.
ownerReputation	Asker’s reputation score at posting, capturing standing and experience beyond raw tenure.

Table 3: Covariate Balance Before and After Matching

Covariate	Unmatched			Matched		
	Tr Mean	Ct Mean	SMD	Tr Mean	Ct Mean	SMD
userTenure	570.95	720.62	−0.167	568.37	544.76	0.026
numQuestionsAskedAT	31.11	26.00	0.059	30.34	28.04	0.027
numHelpProvidedAT	15.27	14.15	0.009	14.36	12.15	0.018
numQuestionsAsked30D	2.00	1.19	0.197	1.96	1.84	0.030
numHelpProvided30D	0.81	0.46	0.068	0.76	0.62	0.028
numQuestionsAsked7D	0.57	0.34	0.174	0.56	0.53	0.023
numHelpProvided7D	0.22	0.12	0.063	0.20	0.16	0.026
tagAcceptShareAvg	0.51	0.44	0.695	0.51	0.51	−0.003
numTags	2.83	3.04	−0.173	2.83	2.83	0.000
bodyLenChars	1289.10	1658.54	−0.183	1286.37	1338.86	−0.026
titleLenChars	53.80	57.50	−0.172	53.79	53.97	−0.009
ownerReputation	4300.57	2541.81	0.115	4130.42	3499.00	0.041

Note. SMD = standardized mean difference. The conventional threshold for acceptable balance is $|\text{SMD}| < 0.1$. All post-matching SMDs fall well below this threshold.

and underscoring the importance of matching on this variable. The propensity model addresses observable selection directly by including posting hour and day, question and title length, and owner reputation (Table 2). After matching, all standardized mean differences fell well below the conventional threshold of 0.1, with the largest residual imbalance below 0.05 (see Table 3). This confirms adequate balance across all matching covariates; the full set is illustrated in the Love plot in Appendix B.

4.5. Analytical Strategy: Cox Proportional Hazards with Time-Varying Covariates

We model the hazard of helping using a Cox proportional hazards model with time-varying covariates. The counting-process formulation of Andersen and Gill (1982) extends the original Cox model (Cox, 1972) to settings with time-varying covariates and recurrent events.

For each matched pair of observations (one treated, one control), we partition the observation window into intervals at the key event times: the start of the window, the posting of the question, and the arrival of the answer (actual for treated users, synthetic for controls).

We specify the hazard function as:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\beta_0 \cdot \text{treatment} + \beta_1 \cdot \text{phasePostQuestion}(t) + \beta_2 \cdot \text{treatedPostQuestion}(t) \right. \\ \left. + \beta_3 \cdot \text{phasePostAnswer}(t) + \beta_4 \cdot \text{isTreatedActive}(t) \right) \quad (1)$$

where $h_0(t)$ is the baseline hazard. *treatment* is a time-constant indicator for treatment group membership (1 if the user’s question received an answer, 0 otherwise); it absorbs stable between-group differences in baseline helping rates. The remaining covariates are time-varying indicators that switch on at the appropriate event times. Because *treatedPostQuestion* switches on when the question is posted and remains on through the post-answer phase, the treated–control gap in the helping hazard equals β_2 during the waiting period and $\beta_2 + \beta_4$ after the answer arrives. The coefficient β_4 is therefore the difference-in-differences estimate of the change at answer arrival: the shift in the treated–control gap at the time of the answer, net of whatever divergence preceded it. We report β_4 as the treatment effect. We report the waiting-period coefficient β_2 separately, as a parallel-trends diagnostic: because no answer has yet arrived during the waiting period, it cannot reflect a response to answer receipt. It measures anticipatory or selection-driven divergence.

To test the moderation by response time (Hypothesis 3), we extend the model with two additional terms:

$$h(t \mid \mathbf{X}(t)) = h_0(t) \exp \left(\boldsymbol{\beta}' \mathbf{X}(t) \right. \\ \left. + \gamma \cdot \text{isTreatedActive}(t) \times \widetilde{\log}(\text{RT}) + \delta \cdot \text{treatedPostQuestion}(t) \times \widetilde{\log}(\text{RT}) \right) \quad (2)$$

where $\widetilde{\log}(\text{RT})$ denotes the standardized log response-time variable (see below), γ captures the moderation of the answer-arrival increment by answer speed, and δ the analogous moderation during the waiting period. The logic mirrors the base model exactly. Just as β_4 alone is the treatment effect, γ alone is the response-time moderation of that effect: it scales the arrival increment β_4 , the difference-in-differences contrast on which the reciprocity effect rests. We therefore test H3 on γ . A negative γ indicates that faster answers strengthen the arrival increment, the direction H3 predicts. The waiting-period moderation δ is the counterpart of β_2 : it scales a pre-answer gap that cannot reflect a response to an answer not yet received. We report it separately as a diagnostic.

Stratification by Tenure.. To test Hypothesis 2, we fit separate models for each of the seven tenure buckets. This allows the baseline hazard and all coefficients to vary freely across levels of user experience, providing a non-parametric view of how reciprocity changes over the user trajectory.

Identification.. Our identification strategy rests on the matched DiD logic embedded in the model. Propensity score matching (Rosenbaum and Rubin, 1983) ensures that treated and control observations come from users with comparable baseline characteristics, including tenure, activity levels, and question topic difficulty. This matching absorbs between-user differences that might otherwise confound the treatment effect.

We do not include user fixed effects in the survival models. Because we match across different users (not within a user), and because the Cox model already leverages within-observation temporal variation through its time-varying covariates (Andersen and Gill, 1982), user fixed effects are not required for identification. The DiD structure compares within-observation changes before and after the answer across treatment groups, isolating the treatment effect from stable individual differences.

The waiting-period covariate (*treatedPostQuestion*) captures any lift in helping that treated users show after asking but before receiving an answer. When this term is near zero, treated and control users move in parallel up to the answer, and the arrival increment β_4 captures the response to receiving an answer rather than to asking a question. When it is not, the two groups diverge before treatment arrives; β_4 still nets out the average waiting-period gap, but we flag such strata and interpret their estimates with caution. Even a near-zero waiting-period term, however, rules out only motivational differences that materialize *before* the answer. We therefore report the waiting-period term explicitly, so that we measure departures from parallel trends, and we return to its magnitude in the Results.

5. Results

5.1. Descriptive Statistics

We analyze 24,349,760 question-level observations from 3,732,706 users on Stack Overflow, half of whose questions received at least one answer (a 50% rate fixed by construction of the matched sample, not a platform base rate). Most users help no one in a given window: they post 0.18 answers to other people’s questions on average (SD = 1.24; median 0), so helping is rare

for any individual user in any single window. Half of the users who got an answer waited less than 0.34 hours for it, though on average users wait a few hours (mean 5.08 hours, SD = 17.06). Our users span every level of platform experience. Most questions are raised by users who had been on the platform one to three years ($n = 6,424,968$), followed by those at one to six months ($n = 4,045,952$) and those in their first week ($n = 4,029,008$).

Table 4: Descriptive Statistics

Variable	Mean	SD	Median	N
Questions (observations)				24,349,760
Unique users				3,732,706
% questions with answer	50.00%			
Help events per window (answers to others)	0.18	1.24	0.00	
User tenure (days)	656	839	328	
Response time (hours, treated)	5.08	17.06	0.34	
<i>Response time (hours, treated) by Tenure Bucket</i>				
< 1 Week	4.51	15.73	0.32	2,014,406
1 Week - 1 Month	3.59	13.83	0.26	775,751
1 - 6 Months	3.83	14.45	0.26	2,022,884
6 - 12 Months	4.24	15.42	0.28	1,491,553
1 - 3 Years	5.11	17.17	0.34	3,212,434
3 - 6 Years	6.69	19.87	0.50	1,807,077
> 6 Years	8.66	22.72	0.75	850,441
<i>N (unique questions) by Tenure Bucket</i>				
< 1 Week				4,029,008
1 Week - 1 Month				1,551,586
1 - 6 Months				4,045,952
6 - 12 Months				2,983,180
1 - 3 Years				6,424,968
3 - 6 Years				3,614,184
> 6 Years				1,700,882

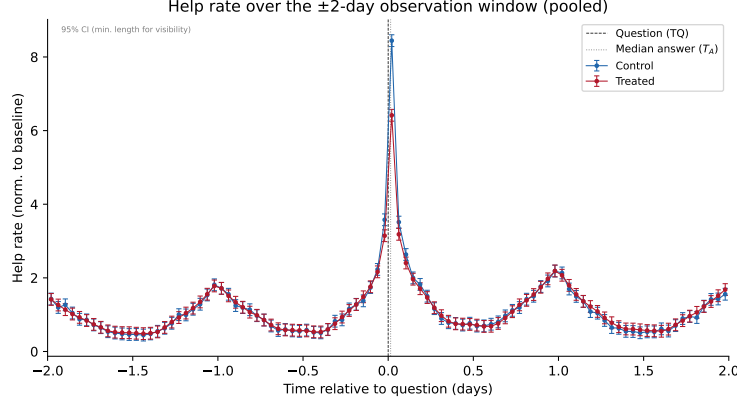
Notes: Help events per window are raw event counts. Estimation applies a one-hour interval rounding (see Implementation Details) under which same-hour events by a user merge, so regression-table *Events* counts (0.1411 per row) sit below this raw mean.

5.2. Descriptive Evidence: Help Rates and the Parallel Trends Assumption

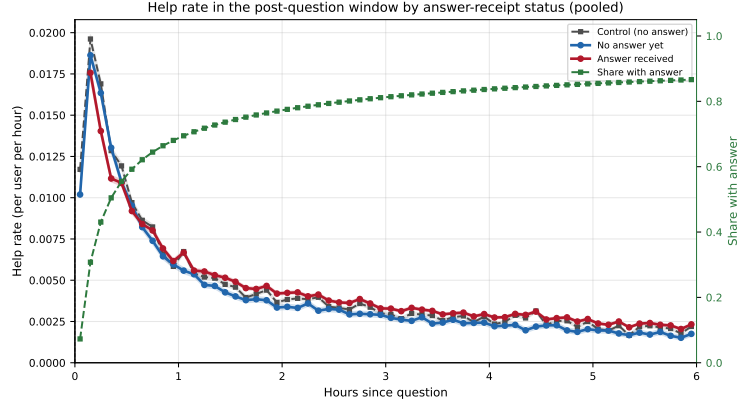
Panel (a) of Figure 3 plots the normalized help rate for treated and control users over the two days before and after question posting. Before the question ($t < T_Q$), treated and control help rates track each other closely, consistent with the parallel-trends assumption and with the effectiveness of propensity score matching. At the moment of posting, both groups' help rates spike sharply: users who have just asked a question are, mechanically, online and active. The spike is more pronounced for controls, who continue waiting for an answer, than for treated users, whose questions are typically answered within the hour. Beyond this mechanical spike, the treated-control differences implied by our estimates (a few percent of the helping hazard) are too small to be visible in raw normalized rates. We return to the control group's larger at-question spike when discussing the response-time pattern.

5.3. Generalized Reciprocity Effect (H1)

To test Hypothesis 1, we estimate the pooled Cox model across all experience levels (Table 5). The treatment effect is the difference-in-differences increment at answer arrival (β_4); we report the waiting-period term (β_2) alongside it as a parallel-trends check. Users who received an answer went on to help others slightly more often than their matched counterparts did, and the gap is small: the arrival increment is $HR = 1.049$ (95% CI [1.044, 1.054], $p < .001$), roughly a 5% higher rate of helping once the answer lands. The waiting-period term is small and negative ($\beta_2 = -0.007$, $p = .01$): distinguishable from zero at this sample size, but an order of magnitude smaller than the arrival increment and opposite in sign. Treated users did not diverge upward from controls before the answer arrived; if anything, they helped slightly less. The arrival increment is therefore not inflated by a pre-answer trend. A waiting-period term this close to zero indicates that asking a question did not, by itself, change how much treated users helped while they waited. Appendix Table F.12 reports matched-pair bootstrap uncertainty for the pooled model, which preserves the matched structure and confirms the model-based interval. Hypothesis 1 thus is supported. The association, however, is smaller than prior observational reports.



(a) Treated versus matched control users, over a fixed ± 2 -day window centered on the question. Red: treated (received an answer). Blue: matched control (did not). Rates are normalized to each group's pre-question baseline; the dashed line marks the question time (T_Q), the dotted line the median answer arrival, and error bars indicate 95% confidence intervals.



(b) Treated users in the post-question window, split by answer-receipt status. Grey dashed: matched controls, who received no answer. Blue: treated users not yet answered. Red: treated users already answered. Green dashed (right axis): cumulative share of treated users answered by each hour.

Figure 3: Raw help rates over the observation window, pooled across all tenure buckets. The two panels answer different questions and are read in different sections: panel (a) bears on the parallel-trends logic of the design (Section 5.2), panel (b) on the re-engagement reading of the response-time pattern (Section 5.5). For legibility panel (a) uses a symmetric question-centered axis ($T_Q \pm 2D$) rather than the answer-anchored analysis window of Figure 1, which closes two days after the answer ($T_A + 2D$) and is the window used in the Cox models. Appendix D reproduces both panels by tenure bucket.

Table 5: Pooled Cox Regressions (All Experience Levels)

	Main	Main + Speed
<i>Treatment effect (difference-in-differences): increment at answer arrival (β_4)</i>		
Answer arrival ($\times$ Post-Answer Received)	0.0479*** (0.0026)	0.0373*** (0.0022)
Hazard Ratio [95% CI]	[1.044, 1.054]	[1.034, 1.042]
Hazard Ratio [bootstrap 95% CI]	[1.032, 1.075]	—
<i>Parallel-trends diagnostic: waiting period (β_2)</i>		
Waiting period (β_2): Received Answer $\times$ Post-Question	-0.0065* (0.0026)	0.0142*** (0.0021)
<i>Response Time Interaction</i>		
Answer arrival $\times$ log(RT) (γ)	—	-0.0636*** (0.0015)
Waiting period $\times$ log(RT) (δ)	—	0.0630*** (0.0013)
N (matched question rows)	24,349,760	24,349,760
Person-interval rows	62,379,420	62,379,420
Events	3,435,973	3,435,973

Notes: Matched-pair bootstrap 95% CI for the pooled arrival increment is from Table F.12.

N counts matched question-level rows (treated + control); the Cox partial likelihood is fit over the person-interval rows expanded from them. The point estimates use the full interval set; only the matched-pair bootstrap replicates (Table F.12) draw an 8M-interval-row subsample per replicate.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; $^\dagger p < 0.1$

In response to the concern that an answers-only outcome may not reflect the full breadth of reciprocity by excluding lower-cost helping acts, Table 6 re-estimates the pooled model under broader outcome definitions. The decomposition confirms that the reciprocity signal is concentrated in *answers*. Comments alone show only a slight elevation (arrival HR = 1.01, 95% CI [1.01, 1.02]), well below the answers-only increment, and adding comments to answers dilutes rather than amplifies the estimate (answers-plus-comments arrival HR = 1.02, [1.02, 1.03]). That is the opposite of what a mechanical undercount of low-cost reciprocity among novices would produce. Edits of other users' posts carry a larger association (edits-only arrival HR = 1.16), but we report them separately because they reflect curation more than costly contribution; the all-channel composite sits between the two (answers-plus-

comments-plus-edits arrival $HR = 1.05$). The waiting-period diagnostic is near zero in every column except edits ($\beta_2 = +0.06$), and the effect enters at answer arrival, mirroring the pooled pattern. The answers-only estimate we use is thus the most pronounced way for users to pay back that does not manifest in low-cost channels with the same magnitude.

Table 6: Reciprocity Effect Decomposed by Help Type

Outcome	Arrival HR [95% CI] (β_4)	Waiting β_2	N	Events
Answers only	1.049 [1.044, 1.054]	-0.01	24,349,760	3,435,973
Comments only	1.013 [1.007, 1.018]	+0.01	24,349,760	2,170,711
Edits only	1.163 [1.155, 1.170]	+0.06	24,349,760	992,369
Answers + comments	1.024 [1.020, 1.029]	-0.00	24,349,760	5,025,482
Answers + comments + edits	1.045 [1.041, 1.050]	-0.00	24,349,760	5,679,602

Notes: Accepting an answer is excluded from the decomposition: it is a self-directed act available only to treated users (a control never receives an answer to accept). Events follow the one-hour interval rounding described in Implementation Details. Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

5.4. Experience Moderation of Reciprocity (H_2)

To test Hypothesis 2, we estimate the model separately within each of the seven tenure buckets (Table 7; Figure 4). The response is strongest at entry and fades to nothing among the most experienced, though the decline across the middle of the tenure range is not strictly monotone. The newest users respond most: in their first week, users who received an answer went on to help roughly 7.1% more often than their matched counterparts (arrival $HR = 1.071$, 95% CI [1.056, 1.086], $p < .001$). The response then dips in the one-week-to-one-month bucket ($HR = 1.03$ [1.01, 1.05]), recovers to a plateau of about 1.05 that holds from one month through three years, and weakens again at three to six years ($HR = 1.02$ [1.01, 1.04], $p < .001$). Only users with more than six years on the platform show no response at all ($HR = 1.00$ [0.98, 1.02], n.s.). Read at this level, the pattern supports Hypothesis 2: reciprocity is a newcomer phenomenon that fades with experience. We do not read the one-week-to-one-month dip as substantively meaningful, and we stop short of describing the decline as strictly graded. Because the newcomer stratum is one of our central findings, we subject its arrival increment to a matched-pair bootstrap that resamples whole pairs and preserves the matched structure. At 500 replicates, the bootstrap 95% CI is [1.06, 1.08], confirming the model-based interval (Appendix Table F.12).

For newcomers, both the arrival increment (β_4) and the waiting-period term (β_2) are positive and statistically significant ($\beta_4 = 0.068$, $\beta_2 = 0.050$, both $p < .001$), whereas among veterans both are near zero. The positive waiting-period term means that newcomers begin helping more even before their answer arrives. The arrival increment nets out that divergence, but a nonzero pre-trend in this stratum is itself a caution: the design cannot fully separate that pre-answer divergence from reciprocity (Section 6.5).

Stack Overflow was substantially disrupted by the arrival of generative-AI tools in late 2022, which sharply reduced human question-and-answer activity (del Rio-Chanona et al., 2024; Burtch et al., 2024). Our own data carry a trace of it: the per-window helping rate is markedly lower after 2020 than before, about 0.057 against 0.164 helping events per matched window. That disruption is not the primary vantage point of this paper, but it invites a natural check: do our findings hold before and after it? Because calendar year enters the design only as an exact-matching stratum, we can re-estimate our two central results, the pooled effect (H1) and the tenure gradient (H2), separately by era (Appendix Table G.13). The pooled effect is stable across the divide: the arrival increment is 1.04 on the pre-2020 subsample and 1.04 on the disjoint post-2020 data (95% CI [1.03, 1.06]), against 1.05 in the full sample. The newcomer effect is not. It is present before the generative-AI era (pre-2020 arrival HR = 1.07, matching the full sample) but not distinguishable from one afterward (post-2020 HR = 1.00, [0.97, 1.03]), on far fewer events (31,814 versus 336,966). We read this as a genuine difference between periods: the pooled association even replicates on data that postdate the disruption, while the newcomer response is well identified in the pre- and early-generative-AI data but appears to weaken after 2020. We develop candidate explanations for why it weakens in the discussion (Section 6.6).

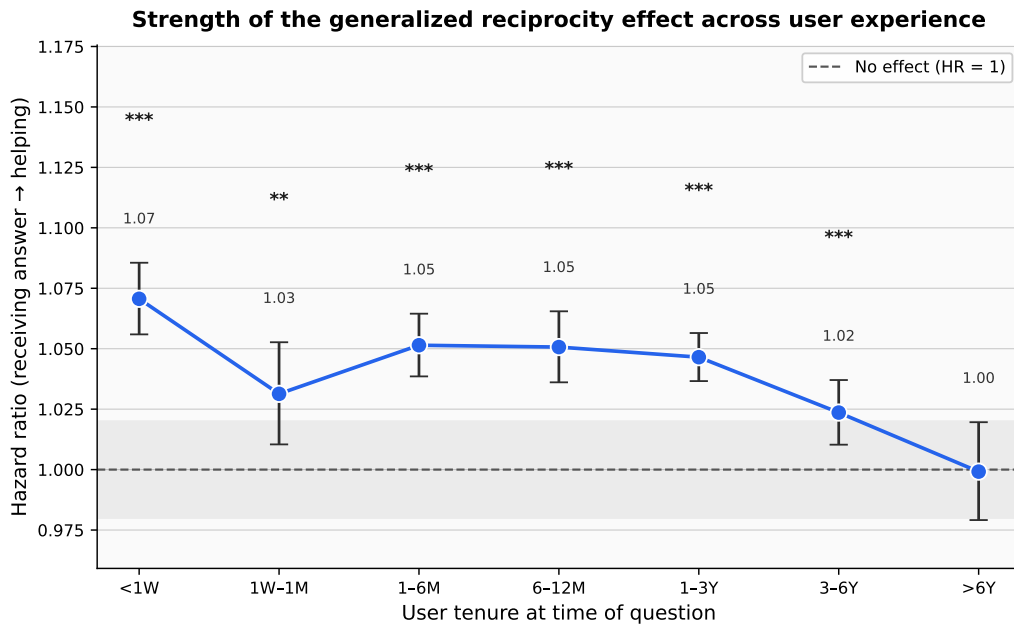


Figure 4: Strength of the generalized reciprocity effect across user experience. Each point shows the arrival-increment hazard ratio (β_4) estimated separately within a tenure bucket; error bars indicate 95% confidence intervals, and the dashed line marks no effect ($HR = 1$). Stars denote significance (** $p < .001$; ** $p < .01$).

Table 7: Cox Regression Results: Effect of Receiving an Answer on Helping Hazard

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Treatment effect: increment at answer arrival (β_4)</i>							
Answer arrival ($\times$ Post-Answer Received)	0.0683***	0.0309**	0.0502***	0.0494***	0.0455***	0.0233***	-0.0008
	(0.0071)	(0.0105)	(0.0063)	(0.0071)	(0.0048)	(0.0067)	(0.0103)
<i>Hazard Ratio</i> [95% CI]	[1.056, 1.086]	[1.010, 1.053]	[1.039, 1.064]	[1.036, 1.065]	[1.037, 1.056]	[1.010, 1.037]	[0.979, 1.020]
<i>Waiting period (β_2, parallel-trends diagnostic)</i>							
Received Answer $\times$ Post-Question	0.0504***	-0.0266*	-0.0147*	-0.0062	0.0000	-0.0107	-0.0167†
	(0.0070)	(0.0105)	(0.0063)	(0.0072)	(0.0049)	(0.0066)	(0.0101)
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	368,780	238,189	694,001	537,263	1,037,470	428,036	132,234

Notes: N = unique questions (treated + control) in the Cox sample. Within each column, treated vs. control counts can differ because tenure is defined per question.

β_4 (the answer-arrival increment) is the difference-in-differences treatment effect; β_2 (waiting period) is reported as a parallel-trends diagnostic and is not added to the effect.

Cox-table standard errors are model-based; matched-pair bootstrap 95% CIs for the pooled arrival increment (β_4) are in Table F.12.

Events = helping events in the full analysis sample (after time rounding). When interval rows exceed the fit cap, estimation uses a matched-pair subsample, but Events still refer to the full sample.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

5.5. Response Time Moderation ($H3$)

To test Hypothesis 3, which predicted that faster responses strengthen the reciprocity effect, we estimate the arrival increment and its response-time moderation separately by tenure bucket (Table 8); Table 9 and Figure 5 give a more flexible discrete-bin view, and the figure sets each baseline bin beside its answer-quality-controlled counterpart. The evidence supports the immediacy prediction. The continuous log-response-time interaction with the arrival increment (γ) is negative in every tenure bucket and statistically significant in all but the most senior: users who waited less for their answer responded more strongly, the direction $H3$ predicts. The moderation is strongest early in the trajectory (about -0.07 to -0.09 from the newcomer through one-year buckets), weakens with tenure (-0.05 at one to three years, -0.02 at three to six), and is indistinguishable from zero among users with more than six years. The discrete bins tell a consistent, if noisier, story. Users who receive help within 30 minutes barely respond at all, while those who receive help between 30 and 60 minutes respond the strongest ($1.19 [1.17, 1.21]$). It declines toward one over the following day (1.06 at one to two hours, 1.02 – 1.05 through one day), with a smaller secondary elevation at one to three days ($1.09 [1.07, 1.11]$). The pronounced early peak shows a sweet-spot pattern rather than a monotone speed gradient; we return to explanations for the sweet-spot pattern below.

The waiting-period gap carries a response-time moderation of its own (δ), positive in every bucket — the counterpart of the β_2 diagnostic, capturing how the *pre-answer* treated–control gap varies with answer speed. Because no answer has arrived yet, we read it as anticipation or selection, not as a response to answer speed. It is separate from γ , the moderation of the arrival increment: the difference-in-differences nets the pre-answer gap out of the increment, so the moderation we report (γ) reflects the change at answer arrival, not treated users’ pre-answer trajectories.

A natural concern is that response time reflects unobserved answer quality. People who answer within minutes tend to post something terse or link-only, whereas those who take longer tend to write fuller, more detailed answers. So, what prompts users to reciprocate may be the quality of the answer they received, not its timing. If quality and response time are correlated in this way, the sweet-spot pattern could reflect quality differences rather than immediacy. We probe the threat with two checks.

Table 8: Response Time Moderation of the Reciprocity Effect

	< 1 Week	1 Wk– 1 Mo	1–6 Mo	6–12 Mo	1–3 Yr	3–6 Yr	> 6 Yrs
<i>Treatment effect: arrival increment (β_4, at mean response time)</i>							
Answer arrival (net post-answer)	0.0101 [†]	0.0178*	0.0348***	0.0417***	0.0501***	0.0353***	0.0139
	(0.0057)	(0.0086)	(0.0052)	(0.0060)	(0.0041)	(0.0056)	(0.0086)
<i>Response-time moderation (γ = arrival; δ = waiting-period diagnostic)</i>							
Answer arrival $\times$ log(RT) (γ)	-0.0704***	-0.0778***	-0.0910***	-0.0863***	-0.0547***	-0.0228***	-0.0015
	(0.0048)	(0.0068)	(0.0039)	(0.0043)	(0.0028)	(0.0036)	(0.0055)
Waiting period $\times$ log(RT) (δ)	0.0236***	0.0725***	0.0892***	0.0829***	0.0670***	0.0453***	0.0278***
	(0.0039)	(0.0060)	(0.0035)	(0.0038)	(0.0024)	(0.0031)	(0.0045)
N	4,029,008	1,551,586	4,045,952	2,983,180	6,424,968	3,614,184	1,700,882
Events	368,780	238,189	694,001	537,263	1,037,470	428,036	132,234

Notes: γ and δ enter response time linearly in standardized log hours; the flexible discrete-bin counterpart of this specification is Table 9 and Figure 5.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; [†] $p < 0.1$

Table 9: Treatment Effect by Response-Time Bin

Response time	HR	95% CI	N	Events
0-15 min	0.998	[0.989, 1.007]	17,412,740	2,484,407
15-30 min	1.004	[0.989, 1.019]	13,934,456	1,812,023
30-60 min	1.190***	[1.174, 1.206]	13,549,736	1,744,784
1-2 hr	1.056***	[1.040, 1.072]	13,211,144	1,698,505
2-4 hr	1.030***	[1.013, 1.047]	12,958,927	1,663,748
4-8 hr	1.017†	[0.999, 1.035]	12,762,008	1,639,963
8-12 hr	1.019	[0.995, 1.044]	12,464,162	1,601,705
12-24 hr	1.048***	[1.028, 1.067]	12,645,932	1,626,049
1-3 days	1.088***	[1.069, 1.108]	12,592,667	1,635,906
>3 days	1.015	[0.992, 1.038]	12,391,574	1,628,425

Notes: Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

The first is the direct test for the response-time pattern: we re-estimate each response-time bin with the three answer-quality controls, which Figure 5 sets beside the baseline fits in panel (b) and Appendix Table G.17 reports in full. This adjustment is deliberately aggressive. Vote score and acceptance are post-treatment, and acceptance in particular requires the asker to return to the platform, so conditioning on them absorbs some of the effect itself, not only the quality confound. The clearest sign of this over-adjustment is that under the controls the fastest bins fall significantly *below* one (0.95), which has no plausible causal reading. Even so, the 30–60-minute peak falls only from 1.19 to 1.03 (95% CI [1.02, 1.04], $p < .001$) and remains the significant early-window maximum, and the one-to-three-day elevation persists (1.09). The sweet-spot pattern remains even after an adjustment likely biased against it.

The second check concerns the pooled *level* of the arrival increment. Appendix Table G.15 reports how it shifts as the same controls enter one at a time. Answer length, which is the least endogenous signal and plausibly correlated with answer quality, barely moves it (from 1.04 to 1.03). The more endogenous vote-score and acceptance signals pull it further, to 1.01 and then below one (0.98); another sensitivity check, where we account for the heavy right tail of vote scores, accompanies these fits (Appendix Table G.18). As above, these level estimates are bounds, not estimates of the effect net of quality: conditioning on post-treatment signals likely takes away from the

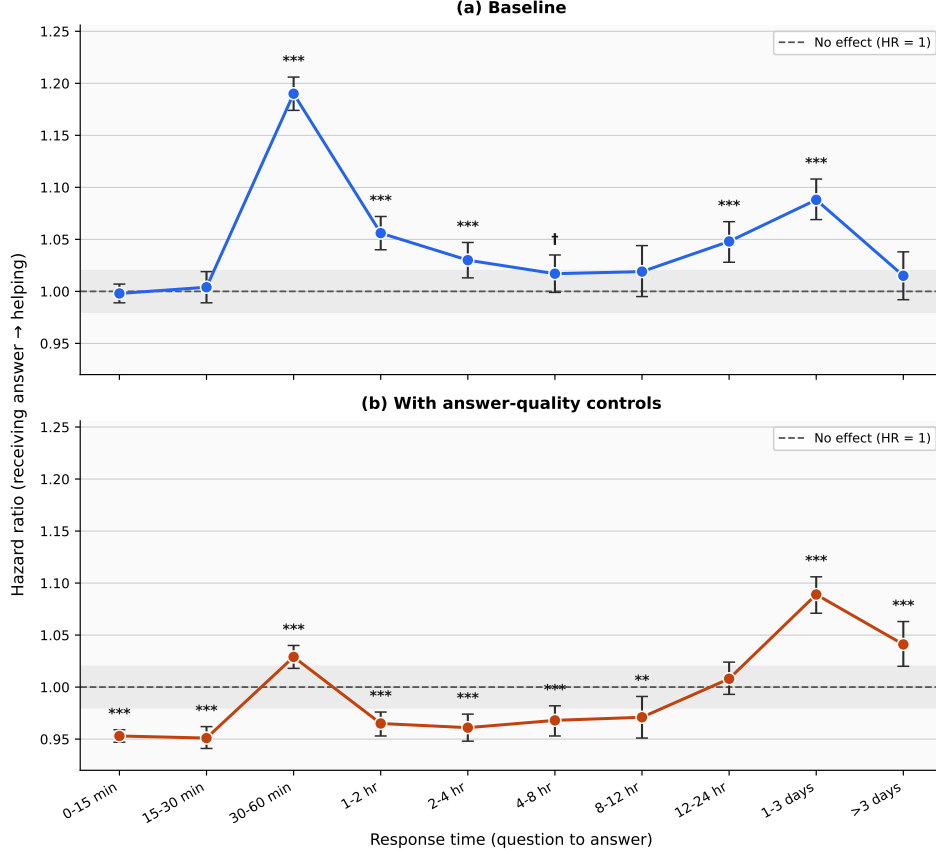


Figure 5: Treatment effect by response-time bin, pooled across all tenure buckets. Each point is the arrival-increment hazard ratio (β_4) estimated within that response-time bin, relative to the matched control; error bars indicate 95% confidence intervals and the dashed line marks no effect ($HR=1$). Panel (a) gives the baseline fits of Table 9. Panel (b) refits every bin with three answer-quality controls, namely acceptance, first-answer score, and answer length, reported in full in Appendix Table G.17. Because the panels share both axes, the adjustment can be read off bin by bin: the 30–60-minute peak shrinks from 1.19 to 1.03 while remaining the early-window maximum, the one-to-three-day elevation is untouched, and the two fastest bins fall below one. Section 5.5 reads those sub-one estimates as a sign that the adjustment is over-aggressive rather than that the effect is absent, since acceptance and vote score are post-treatment.

effect itself. The genuine residual caveat is that our quality measures are coarse, so we cannot cleanly separate the effects of timing and quality, and cannot rule out that answer quality, imperfectly measured, remains a rival to a timing-based reading of the response-time pattern.

One interpretation of the sweet-spot pattern is *re-engagement*: users who post a question may remain actively engaged while waiting during the ongoing session on the platform, elevating helping during the waiting period regardless of whether an answer arrives. On this account, a fast answer might lead users to end their session and, with it, their engagement without returning to the platform. If an answer arrives after they have ended their session, they might return to review it and reciprocate in the newly started session triggered by the answer. Later answers are more likely to re-engage the user in another session, which answers in the same initial session do not.

Panel (b) of Figure 3 provides descriptive context, comparing users who have received an answer (red) with users who will receive one but are still waiting (blue), alongside matched controls. In the first half-hour after the question — when only the fastest answers have arrived — users still waiting help at the highest rate, consistent with the idea that users help while waiting on the platform. From roughly an hour onward, the ordering flips: users who have received an answer help at a modestly higher rate than those still waiting and than controls, consistent with a small post-answer elevation. We read this as descriptive evidence for the re-engagement mechanism. Appendix D disaggregates the same comparison by tenure bucket and shows the same qualitative pattern in every stratum. These raw-rate orderings are descriptive: the differences are small, and answer-arrival timing is not random within the treated group, so we do not read them as decisive evidence for any one mechanism. Still, they are consistent with the modest reciprocity effect and the re-engagement mechanism the models estimate.

Overall, however, re-engagement is speculative since we do not observe session boundaries (logins, logouts, inactivity gaps). We present it as one candidate among several, not as an established mechanism. We return to this question in Section 6.3.

6. Discussion

6.1. Methodological Contribution

Researchers who study generalized reciprocity from time-series platform data face a persistent difficulty: generally active users are more likely to

ask questions and post answers. A correlation between help received and help given can therefore reflect nothing more than how active a user is. Prior observational studies that report positive associations without accounting for this selection (Baker and Bulkley, 2014; Chen et al., 2018; Joyce and Kraut, 2006) may have overstated the role of reciprocity.

Our matched difference-in-differences survival design addresses that confound. It compares changes in helping, before and after a question is posted, between users whose questions received answers and propensity-score-matched users whose questions did not. The staggered adoption design and modeling the waiting period separately allow us to identify the generalized reciprocity effect associated with answer arrival. The estimate this yields is small and strongest among newcomers, and it is substantially smaller than many earlier reports, which is what activity confounding in those designs would produce.

The design travels past this particular question. Any platform event that arrives at a time the researcher does not control — a first review, a moderator intervention, a badge, a reply — raises the same confound, and the same comparison of matched users before and after the event addresses it. We therefore offer the design, and not only the estimate, as a contribution that researchers and platform designers can use to study behavioral interdependencies between different user-driven platform events.

6.2. Reciprocity as a Contributor-Recruitment Mechanism

Users respond to being helped less the longer they have been on the platform. The newest respond most ($HR = 1.07$) and the most experienced not at all (arrival $HR = 1.00$; Table 7). Read against the theory, this gradient does more than qualify the reciprocity account. It relocates it. Generalized reciprocity on the platform we study is not the general driver of cooperation among established members that the literature often takes it to be. It is a channel that works at the point of entry and weakens as users settle in.

Why it closes is what the theory has to explain. The norm-displacement account developed in Section 2.3 offers one answer: as users gain experience, the community’s own incentives crowd out a general reciprocal impulse. Experienced users on Stack Overflow have reasons to help that newcomers do not yet have — reputation, visible status, a stake in the community. That the most experienced users do not respond at all fits this account. Our design tracks what users did rather than what moved them, so we cannot confirm it.

The reframing also makes sense of an inconsistent literature. Researchers who sample mostly experienced users, or who pool across experience levels dominated by experienced contributors, should find weaker reciprocity effects. Those who catch users early should find stronger ones. Prior findings may diverge, then, not because anyone erred, but because reciprocity is not a fixed property of a population.

What the gradient does not tell us is what takes reciprocity’s place. Our analysis establishes that generalized reciprocity declines with experience; it does not characterize the motives that replace it. The theoretical argument implicates community-specific norms and status concerns, but testing that account needs direct measures of them — surveys of experienced contributors, or experiments that manipulate reputation visibility and observe whether helping follows.

6.3. The Response-Time Pattern and Its Interpretation

The response-time analysis tested a prediction drawn from laboratory studies of gratitude: that faster help produces stronger reciprocity. That much fits an emotional-immediacy account, in which help arriving while the need is still felt draws a stronger response. The evidence supports it across the user trajectory. We also find, however, that users who receive an answer immediately (within 30 minutes) show no reciprocity. Instead, users who wait for their answers a little show the strongest reciprocity. A pure immediacy account gives no reason why extremely fast answers should do nothing at all.

Three readings of that anomaly compete, and our data separate none of them cleanly. The first is answer quality: the quickest replies may also be the thinnest, which the Results adjudicate as far as the dump allows (Section 5.5). The second is re-engagement: an answer that lands while a user is still on the site may end their visit, while one that lands later brings them back. The third is question characteristics: response time stands in part for properties of the question that the tag-level accepted-answer rate does not capture, since questions that draw fast answers may differ from those that wait. We advance all three as conjectures. We observe neither session boundaries nor question difficulty, and our quality signals are coarse.

The pattern can also speak to the norm-displacement account. Among the most experienced users, the arrival increment and its response-time moderation are both near zero, which is consistent with the mechanism having faded rather than merely weakened. While this needs to be treated with caution, this finding gives further credibility to our theoretical account.

Settling the interpretation needs data the archive does not hold. Session-level records would test re-engagement directly. Sophisticated, text-based measures of question complexity or specificity would separate timing from difficulty. Randomly assigned answer timing would separate both from immediacy. Each of these may be available to a platform or an experimenter, offering the opportunity to build on this work.

6.4. Implications for Platform Design

Because the response we observe is modest and strongest among newcomers, designers can use it mainly to recruit new contributors rather than to keep established members contributing.

Designers who want to grow their contributor base should therefore make sure that new users get answers to their questions. Newcomers who received an answer went on to help about 7% more often than their matched counterparts ($HR = 1.07$; Table 7), or roughly 8 additional helping events per 1,000 answered newcomer questions (Table 10). The absolute figure is small because helping is rare in this population to begin with: matched newcomer controls average about 0.11 helping events per observation window, and most control observations record none at all. Put differently, about 125 newcomer questions must be answered to produce one additional act of helping. This is still the highest yield of any tenure group, next to the statistically indistinguishable one-to-twelve-month strata (roughly 126–128 questions) but clearly more favorable than the pooled figure of about 158. The case for concentrating on newcomers therefore rests both on their being the group that responds most and on a per-answer yield at least as favorable as any other group’s. The most experienced users do not respond (arrival $HR = 1.00$), so answering their questions buys a designer nothing. Designers thus have most leverage with reciprocity at the point of entry.

Table 10: Absolute Effect Sizes and NNT

Row	Control rate	HR	ARD	Events/1k	NNT	NNT CI
< 1 Week	0.1133	1.071	0.00800	8.00	125.0	103.2–157.8
1 Week - 1 Month	0.1401	1.031	0.00439	4.39	227.8	135.5–684.6
1 - 6 Months	0.1542	1.051	0.00793	7.93	126.1	100.6–168.2
6 - 12 Months	0.1542	1.051	0.00782	7.82	127.9	99.0–179.6
1 - 3 Years	0.1370	1.047	0.00637	6.37	157.0	129.2–199.4
3 - 6 Years	0.1032	1.024	0.00244	2.44	410.7	261.5–939.9
> 6 Years	0.0716	0.999	-0.00006	-0.06	—	—
Pooled (all tenures)	0.1287	1.049	0.00631	6.31	158.4	142.9–177.5

Notes: Based on the answer-arrival increment (β_4), the headline estimand. *Control rate* is the mean helping events per observation window among matched control (unanswered) questions — a recurrent-event rate, not a risk. $\text{ARD} = \text{control rate} \times (\text{HR} - 1)$ treats the hazard ratio as a rate ratio (a first-order approximation); $\text{Events}/1k = 1000 \times \text{ARD}$; and $\text{NNT} = 1/\text{ARD}$ is an approximate “answers needed per additional helping event,” not a true number-needed-to-treat.

Response speed is a weaker basis for design than it first appears. Users who received an answer more quickly generally responded more strongly, and those answered between 30 and 60 minutes responded the most (Tables 8 and 9). The obstacle is not the size of the pattern but its interpretation: it admits two readings that point designers in different directions. We therefore hand designers two questions to run on their own platforms rather than a specific intervention.

The first question is how answers land in a user’s session. Platforms record what our data do not: when a user arrives, how long they stay, and whether an answer reaches them inside that visit or after it has ended. If the sweet spot reflects re-engagement, an answer that arrives mid-visit competes with the activity a user is already engaged in, while one that arrives after the visit gives them a reason to come back. Designers would then be working on the return trip rather than the clock. What matters is when the notification lands, whether the answer is still visible when the user next opens the platform, and how prominently it is surfaced. Time to first answer would be the wrong thing to optimize.

The second question is whether fast answers are *low-cost contributions*: replies written in minutes that are terse or link-only, and that clear the queue without doing much for the asker. The two fastest bins are indistinguishable from no effect (0.998 and 1.004; Table 9), which is what we would expect if the quickest replies are also the thinnest. A newcomer whose first exchange

with the community is a curt reply may simply not return. If low-cost contributions drive the fastest bin, the design problem sits in quality at the fast end rather than in delay, and platforms would do better to discourage quick throwaway replies to newcomers than to shorten the queue further. Our own checks push against this reading: the 30–60-minute peak survives an aggressive quality adjustment (Section 5.5). The quality signals in the public dump are too coarse to settle the fast end outright, but platforms that have high-fidelity measures of answer quality and newcomer retention can settle it directly.

Both readings lead to the same caution. Designers stand on firmer ground making sure that newcomers’ questions get answered at all, and that users notice when they do, than tuning for a particular response window.

6.5. *Limitations and Future Research*

Four questions mark the boundaries of this study and, with them, the openings for future research: what we measure, what we can identify, what we cannot see, and where the estimates apply. We take each in turn.

Start with what we measure. Our helping outcome is the posting of an answer to another user’s question, which we treat as a costly public contribution. Reciprocity can also run through channels we do not observe. On the platform, upvotes are the clearest case: the public dump withholds the voter’s identity, so votes given cannot be attributed to a user at all. Off the platform, a helped user might answer a colleague directly, write the solution up elsewhere, or pass it on somewhere Stack Overflow never records. Table 6 sets out which channels the outcome does and does not include. If newcomers reciprocate through these unobserved channels more than veterans do, our estimates miss more of their reciprocity than of the veterans’, which makes the reported tenure finding a conservative one.

Turn next to what we can identify. Our estimate of reciprocity is the arrival increment, the change in helping at the moment an answer arrives. It is identified under several assumptions, most critically, that nothing else which raises helping changes at that same moment.

Three kinds of selection bear on that assumption. The matching handles selection on *observables*, the characteristics that predict getting answered, and leaves every covariate below conventional balance thresholds (Figure B.7). The difference-in-differences removes selection on *stable unobservables*, a lasting disposition to both attract answers and give help, since each user is measured against their own baseline. The third kind remains:

something that changes about the user inside the window and affects answer arrival and subsequent helping at once. Someone who becomes more active may draw an answer sooner and, for the same reason, help more. We think this assumption is defensible because the waiting-period term catches everything that already surfaces before the answer.

However, if any behavioral change (that caused the answer) affects helping only after the moment of answer receipt, we cannot separate it from reciprocity. Fully settling it needs a design that makes answer arrival exogenous: a field experiment that seeds answers to a random subset of questions, or a platform change that shifts answering capacity for reasons unrelated to who is asking.

That brings us to what we cannot see. The design captures a behavior change but not the psychology behind it. Gratitude, moral obligation, social expectation, incentive displacement, and re-engagement are all *proposed* interpretations, each consistent with part of the pattern and none identified by our data. If social expectation is the behavioral mechanism, users who feel more visible after being answered should be helping more. However, the change in visibility (and the feeling of being observed) is not obvious. Being answered lifts a question in activity-sorted feeds for a while, but it also takes the question out of the unanswered queue that potential answerers browse. Testing visibility as a mechanism needs a measure of how visible a question was at the time, which the dump does not provide. Beyond that, we observe no session boundaries, no reputation salience, and no emotional states. Which of gratitude, moral obligation, and social expectation actually moves users in a setting like this one is not a question our data can settle, and we leave it to psychologists with the instruments to measure it. Separating these pathways calls for behavioral records paired with measures taken while people are on the platform: experience sampling of emotional state, or direct measures of how users reason about obligation when they return to a thread.

Finally, where the estimates apply. Stack Overflow is a large, relatively anonymous, gamified platform built around technical question-answering. The patterns we estimate may not transfer to smaller or relation-based communities such as enterprise Slack channels, academic peer review, or intimate hobby forums, where social ties and non-status motives differ. We therefore refrain from claiming that generalized reciprocity operates on platforms with different dynamics and mechanisms of social engagement. Our estimates describe one platform, and largely its pre-generative-AI era.

Matching narrows the scope a second time, and in a way worth stating

plainly. Our exact strata are calendar year, primary tag, and number of tags, so a treated question enters the estimate only if an unanswered question exists in the same stratum. Roughly half the unanswered pool has no on-support match, and the control questions that do survive are drawn disproportionately from more readily answered tags. What we estimate is therefore the effect for answered questions in mainstream, well-trafficked tag areas. It is not the effect for the long tail of niche or rarely answered topics, and whether reciprocity operates differently in that tail is a question this design cannot reach. The restriction does run in a conservative direction, since the retained controls are the more answerable ones and so set a higher comparison baseline. It is also not concentrated in the newcomer stratum that carries our central result, which is dropped at close to the arm-wide average rate. Appendix C reports the counts.

6.6. Generalized Reciprocity in the Generative-AI Era

Our analysis window runs from Stack Overflow’s inception through 1 April 2025, and it should be read as a *historical snapshot* rather than a guide to contemporary platform design. Because calendar year is an exact-matching stratum, the pre-2020 period supplies most of the analysis, which dates our results to a platform where community answering was still the norm. The era split reported in Section 5.4 sharpens the point: the pooled effect survives the divide, while the newcomer response does not. The recruitment channel we identify, in other words, weakened at about the point the platform began to change.

Why it weakened is a question our data cannot answer, and we offer two conjectures rather than a conclusion. One is composition: the newcomers who still join after ChatGPT’s release are a selected group, and the post-2022 exit fell hardest on the junior users who carried the earlier response (Burtch et al., 2024). The other is opportunity: activity declined most in the topics where generative AI tools perform best (del Rio-Chanona et al., 2024), so the questions that remain may be harder, leaving a newly helped newcomer with fewer questions they can answer. Both would weaken the recruitment channel for reasons outside the reciprocity mechanism itself. Neither is testable here, since we observe neither question difficulty over time nor who chose not to join.

The larger question is what becomes of reciprocal recruitment when a machine supplies the answer. If newcomers increasingly get help from an assistant rather than a person, the human exchange that gave them their

modest push into contributing thins out, and it thins out exactly where we find the mechanism working. The demand for human contribution does not disappear with routine question-answering, however. Systems still depend on people for correction, feedback, and the ground-truth detail that no model generates on its own. Whether generalized reciprocity can recruit those contributors when the help that triggers it comes from a machine is a question this design can be re-run to answer as the data accumulate.

7. Conclusion

Generalized reciprocity is widely credited with sustaining cooperation on knowledge-sharing platforms, yet observational tests run into a stubborn confound: generally active users ask and answer more questions at the same time. Our matched difference-in-differences survival design confronts that confound directly and delivers a cleaner test of whether people help more once their own questions are answered. The response to being helped is strongest among newcomers and fades to nothing among the longest-tenured users. On Stack Overflow, generalized reciprocity works less as a general engine of cooperation among established members than as a channel for recruiting contributors at the point of entry. This continues to matter as generative-AI systems absorb routine question-answering. People still supply what machines cannot generate on their own: feedback, input, correction, and the ground-truth knowledge that populates shared resources, such as local detail added to mapping platforms. For much of this, these systems in fact depend on the people who provide it. Understanding how such contributors are recruited therefore grows more important, not less. Our study contributes both a finding and a framework: the recruitment leverage of generalized reciprocity is greatest at the point of entry, and our design offers a re-runnable way to measure and strengthen it wherever human contribution still matters.

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Ethics statement

This study analyzed de-identified, publicly available archival records from Stack Overflow. No interaction with human participants occurred, and no

personally identifiable information was collected or analyzed. Ethical approval was not required under applicable institutional policies for secondary analysis of public platform data.

Data availability

Stack Overflow public data dumps were obtained from the Internet Archive (<https://archive.org/details/stackexchange>). Analysis code and step-by-step instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review). Processed analytical datasets used in the matching and survival analyses are available from the corresponding author upon reasonable request.

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Appendix A. Replication materials

Analysis code and instructions to reproduce all tables and figures are available in a public repository (URL withheld for blind review).

Appendix B. Matching Diagnostics

Figure B.6 shows sufficient common support between the treatment and control groups: the propensity-score distributions of the original treated and control groups are similar, and the distribution of the matched control group is nearly identical to that of the treated group.

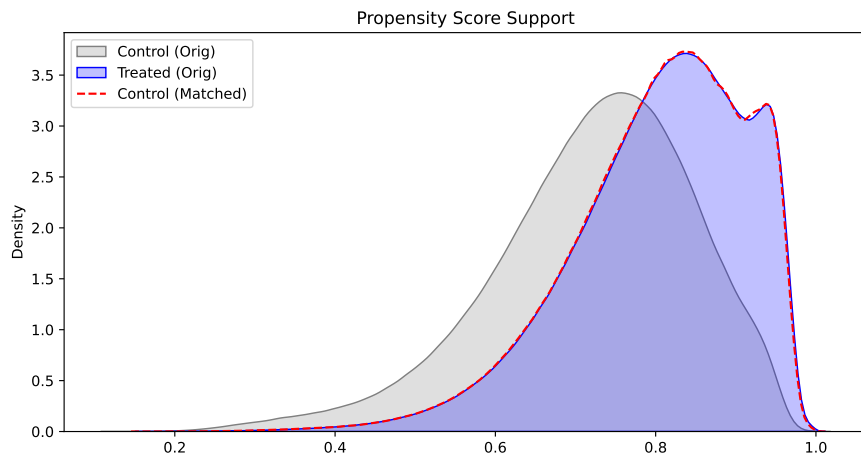


Figure B.6: Distribution of propensity scores of treatment group, control group, and matched control group

Figure B.7 displays the standardized mean differences (SMDs) for the matching covariates before and after matching. Several covariates show noticeable imbalance before matching; after matching, all SMDs are close to zero.

Appendix C. Common Support and the Scope of the Estimand

Matching restricts the scope of the estimand rather than the set of users we observe. Of the 4,999,330 unanswered questions in the pre-matching sample, 2,481,556 serve as matched controls. The rest either lack a valid exact stratum or complete matching covariates, or are never selected as a nearest neighbor within the caliper, so roughly half the unanswered pool has no on-support match. Because the exact strata are calendar year, primary tag, and number of tags, the control pool that survives is drawn from more readily answered tags than the discarded tail.

The reported contrast is therefore the effect for answered questions that have an on-support unanswered counterpart. This is a generalizability restriction inherent to matching, and a conservative one: because the discarded questions are the less answerable ones, the more-answerable retained controls set a higher, not lower, comparison baseline. The discarding is also not concentrated in the newcomer stratum that carries our central result, where the

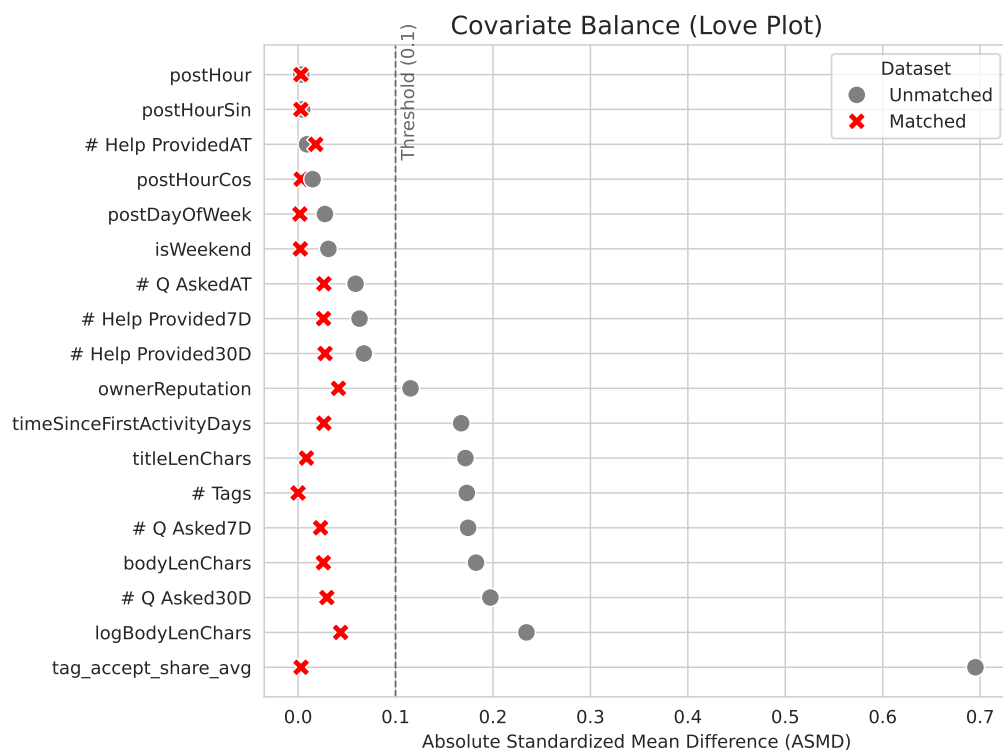


Figure B.7: Visualization of balance in the unmatched and matched dataset

< 1 week bucket is dropped at close to its arm-wide average rate.

Appendix D. Help Rates Over the Observation Window by Tenure

This appendix reproduces both panels of Figure 3 separately by tenure bucket. Figure D.8 disaggregates the treated–control comparison of panel (a), and Figure D.9 the answer-receipt comparison of panel (b).

Figure D.8 shows that treated and control help rates track each other closely before the question in every tenure bucket, supporting the parallel-trends logic within each stratum. The sharp spike at question time is common to both groups and is somewhat larger for controls; the small post-answer elevations estimated in Table 7 and Figure 4 (hazard ratios of a few percent) are below what these raw normalized rates can visually resolve.

Figure D.9 shows the ordering discussed in Section 5.5 holding in every tenure stratum. Users still waiting for an answer help at the highest rate in the first half-hour after the question, and from roughly an hour onward users who have already received an answer help at a modestly higher rate than both those still waiting and the matched controls. As in the pooled panel, these are raw rates rather than estimates: the differences are small, and answer-arrival timing is not random within the treated group.

Appendix E. Estimation Sample Sizes

Table E.11 reconciles the several sample sizes reported across the Cox tables. Each matched pair contributes one treated and one control question-level row (a control reused across pairs is counted once per pair it anchors), and the Cox partial likelihood is fit over the person-interval rows expanded from those matched rows. The pooled point estimates use the full interval set; only the matched-pair bootstrap replicates (Table F.12) cap each replicate at approximately 8 million interval rows. Users and questions are dataset-wide distinct counts and do not partition by tenure bucket, so the bucket rows report only the matched-row, interval-row, and event counts that do.

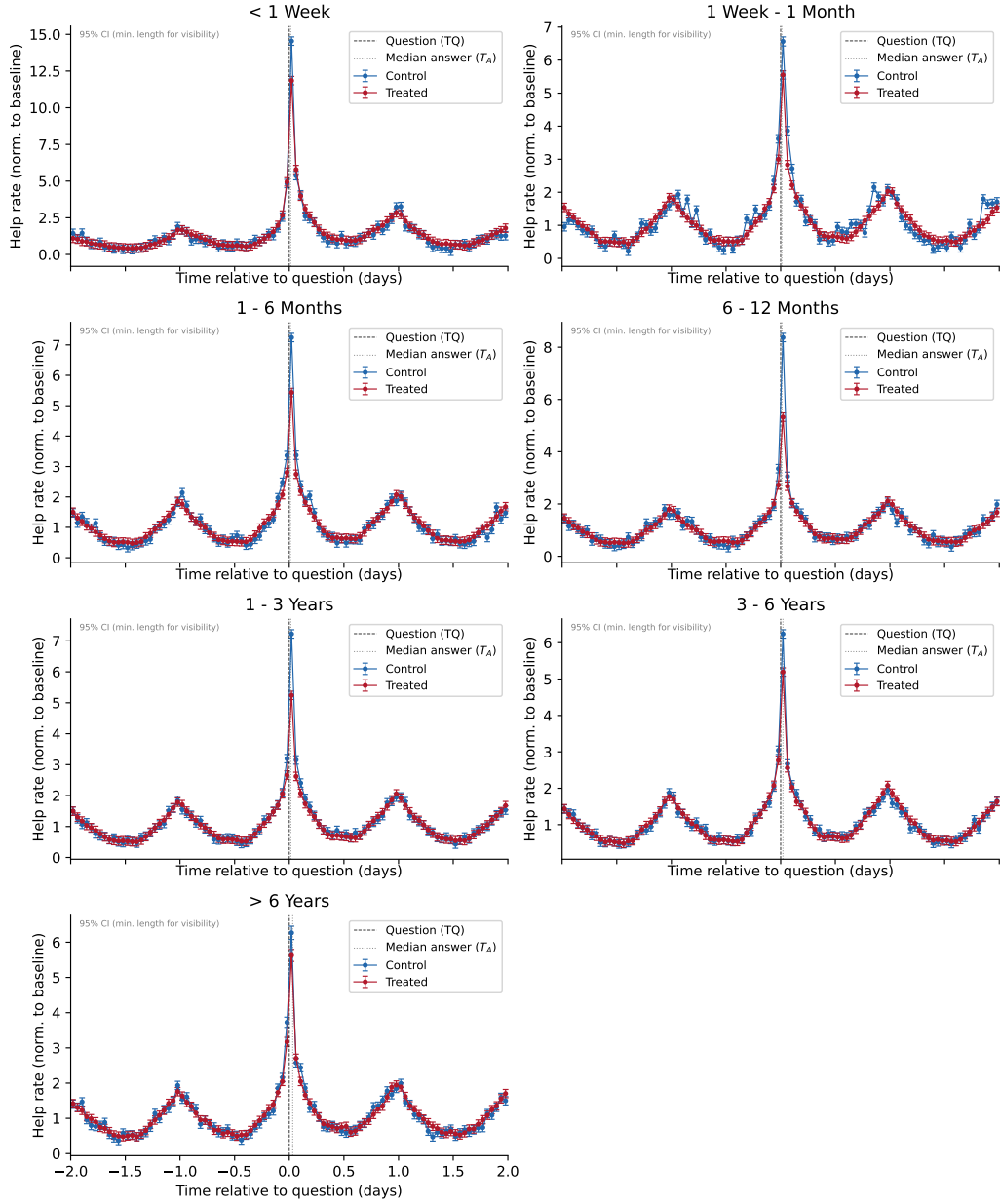


Figure D.8: Help Rate Over the Observation Window, by Tenure Bucket. Each panel plots the normalized help rate (relative to each group's pre-question baseline) for treated users (red) and matched control users (blue) over the ± 2 -day window around the question-posting time (T_Q , dashed vertical line). The dotted vertical line marks the median answer arrival time (T_A). Error bars indicate 95% confidence intervals.

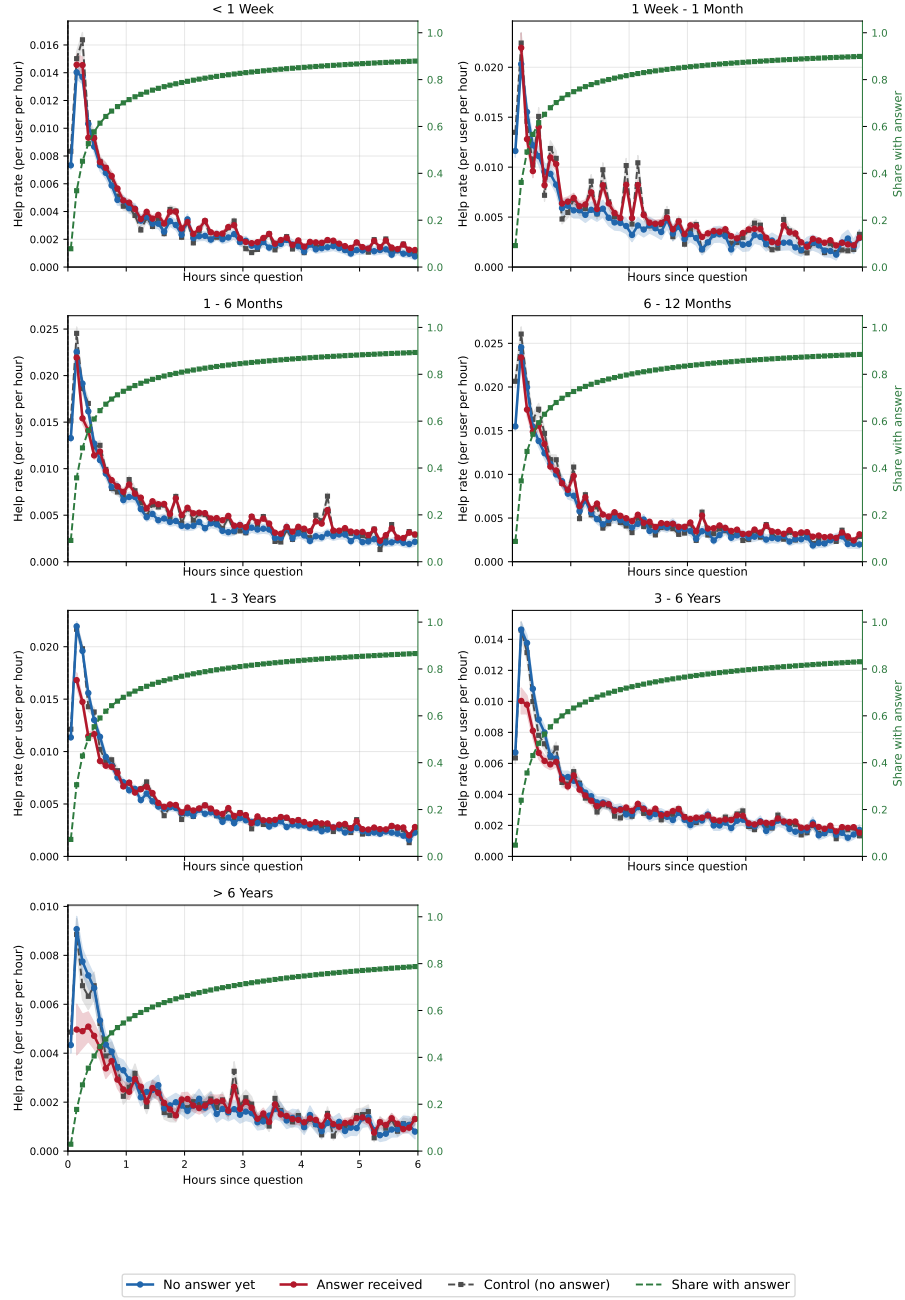


Figure D.9: Help rate in the post-question window by answer-receipt status, separately by tenure bucket. Grey dashed: matched control users who received no answer. Blue: treated users who have not yet received an answer (No answer yet). Red: treated users who have already received an answer (Answer received). Green dashed (right axis): cumulative share of treated users who have received an answer by each hour. This is panel (b) of Figure 3 disaggregated by tenure.

Table E.11: Estimation Sample Sizes by Model

Model	Users	Questions	Matched rows	Interval rows	Events
Pooled (Models A & B)	3,732,706	14,656,436	24,349,760	62,379,420	3,435,973
<i>Model A, by tenure bucket</i>					
< 1 Week	—	—	4,029,008	10,055,987	368,780
1 Week - 1 Month	—	—	1,551,586	3,895,000	238,189
1 - 6 Months	—	—	4,045,952	10,248,285	694,001
6 - 12 Months	—	—	2,983,180	7,632,877	537,263
1 - 3 Years	—	—	6,424,968	16,588,890	1,037,470
3 - 6 Years	—	—	3,614,184	9,444,885	428,036
> 6 Years	—	—	1,700,882	4,513,496	132,234

Notes: Matched rows are treated + control question-level rows in the Cox sample (= $2 \times$ matched pairs); a control reused across pairs contributes one row per pair it anchors. Interval rows are the person-interval rows expanded from the matched rows and fit by the Cox partial likelihood; the point estimates use the full interval set, while the matched-pair bootstrap (Table F.12) caps each replicate at 8M interval rows. Users and Questions are dataset-wide distinct counts and do not partition by tenure bucket (a user’s questions, and reused control questions, can fall in several buckets), so the bucket rows leave those columns blank. Events = helping events in the full analysis sample (after time rounding).

Appendix F. Bootstrap Inference and Absolute Effects

Table F.12 reports matched-pair bootstrap 95% confidence intervals for the Model A treatment effect — the difference-in-differences increment at answer arrival (β_4) — for the pooled sample (100 replicates) and, because it carries the paper’s headline result, the newcomer (< 1 week) stratum at 500 replicates. Absolute effect sizes (absolute rate differences and the implied answers-needed-per-additional-helping-event) are reported in the main text (Table 10, “Implications for Platform Design”).

Table F.12: Matched-Pair Bootstrap Uncertainty for the Answer-Arrival DiD Increment (β_4 ; Model A)

Scope	HR	Bootstrap 95% CI	Replicates	N
Pooled	1.049	[1.032, 1.075]	100/100	24,349,760
< 1 Week	1.071	[1.057, 1.084]	500/500	4,029,008

Notes: Replicates resample whole matched pairs with replacement. HR is the answer-arrival increment $\exp(\beta_4)$, the difference-in-differences treatment effect.

Appendix G. Robustness: Cohort Heterogeneity, Observable Selection, and Answer Quality

Table G.13 addresses the concern that pooling entry cohorts from 2008 through 2025 conflates structurally different eras of the platform (reputation inflation, tag evolution, and the post-2022 generative-AI disruption). Because calendar year enters the matching design only as an exact stratum, treated and control questions within a pair always share a year; to probe heterogeneity *across* years we re-estimate Model A on the pre-2020 subsample, both pooled and in the newcomer (< 1 week) stratum. The pooled pre-2020 arrival hazard ratio is 1.04 (versus 1.05 in the full sample), and the pre-2020 newcomer estimate reproduces the full-sample figure (arrival HR = 1.07 in both), as expected since the pre-2020 rows are most of the sample, indicating that the tenure gradient was already present before the generative-AI era. Because calendar year enters as an exact-matching stratum, the pre-2020 subsample is not disjoint from the full sample but a large sub-population of it, retaining 79% of matched rows and 91% of helping events (Table G.13). This check is therefore compositional — it dates our estimates to the pre-generative-AI platform — rather than an independent replication on separate data; the disjoint post-2020 subsample is reported in the main text, where the pooled effect persists but the newcomer effect does not.

Table G.13: Cohort Heterogeneity Robustness

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
Full sample, pooled	1.049	[1.044, 1.054]	-0.007*	24,349,760	3,435,973
Full sample, newcomer (< 1 week)	1.071	[1.056, 1.086]	+0.050***	4,029,008	368,780
Pre-2020, pooled	1.037	[1.031, 1.042]	-0.000	19,187,320	3,144,155
Pre-2020, newcomer (< 1 week)	1.068	[1.052, 1.085]	+0.071***	3,155,176	336,966
Post-2020, pooled	1.043	[1.029, 1.057]	+0.011†	5,162,440	291,818
Post-2020, newcomer (< 1 week)	1.001	[0.968, 1.034]	-0.026	873,832	31,814

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table G.14 reports the pooled Model A after adding pre-treatment observable selection controls (posting hour and weekday, number of tags, body and title length, and asker reputation). View count is *not* included: it accrues after posting and is higher for answered questions, so conditioning on it would open a post-treatment / collider channel already excluded from the propensity score. The pooled arrival increment remains essentially unchanged once

these pre-treatment observables are held fixed (arrival HR = 1.04, versus 1.05 in the baseline specification).

Table G.14: Observable Selection Controls

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
Baseline (Model A)	1.049	[1.044, 1.054]	-0.007*	24,349,760	3,435,973
+ observable controls	1.040	[1.036, 1.044]	+0.002	24,349,760	3,435,973

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table G.15 reports the pooled speed-specification arrival increment as the three answer-quality controls (first-answer length, vote score, and acceptance) are entered one at a time. The gradient is informative about what each signal absorbs: answer length — the least endogenous signal, fixed by the answerer at composition — barely moves the estimate (1.038 baseline to 1.032); vote score brings it to 1.010; and the full specification, adding acceptance, pushes it below one (0.982), with the same pattern in the newcomer stratum (0.95; Table G.16). Because vote score and, especially, acceptance are post-treatment and endogenous to the outcome (accepting requires the asker to return to the site, the same re-engagement that produces helping), these level estimates are a bounding exercise rather than a causal decomposition: the association is robust to the terse-versus-detailed concern (length), and it is reversed only by the signals a bad-control logic predicts would over-adjust.

Table G.15: Answer-Quality Robustness

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
Speed specification, baseline	1.038	[1.034, 1.042]	+0.014***	24,349,760	3,435,973
+ answer length	1.032	[1.027, 1.036]	+0.006**	24,349,760	3,435,973
+ answer score	1.010	[1.006, 1.015]	-0.015***	24,349,760	3,435,973
+ length, score, and acceptance	0.982	[0.978, 0.986]	-0.046***	24,349,760	3,435,973

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table G.16 reports the newcomer (< 1 week) bucket under several specifications. The baseline arrival increment is 1.07. Adding the same pre-treatment observable controls — still *without* view count — raises the arrival increment to 1.13, which indicates the newcomer association is not an artifact of *observable* selection. This gain warrants caution rather than reassurance, however: the waiting-period diagnostic rises in step, from $\beta_2 =$

+0.050 to +0.122, essentially the magnitude of the arrival log-increment itself ($\log 1.13 = 0.121$). Under these controls, treated newcomers already diverge upward from their matched controls by nearly the full effect size *before* the answer arrives, so the two terms inflate together. We therefore read the specification as ruling out observable selection as the sole driver, not as independent corroboration of the arrival increment; it also sharpens the parallel-trends caveat that already attaches to this stratum. Adding answer-quality controls, by contrast, pushes the newcomer increment below one (arrival 0.95), but this mirrors the pooled pattern and, as above, reflects over-adjustment on post-treatment signals (vote score and acceptance) rather than a quality confound the design fails to resolve.

Table G.16: Newcomer Bucket Robustness (< 1 Week)

Specification	Arrival HR	95% CI	Waiting β_2	N	Events
Model A, baseline	1.071	[1.056, 1.086]	+0.050***	4,029,008	368,780
Model A + observable controls	1.129	[1.117, 1.142]	+0.122***	4,029,008	368,780
Model B (speed), baseline	1.010	[0.999, 1.021]	+0.008	4,029,008	368,780
Model B + answer length	1.003	[0.992, 1.014]	-0.001	4,029,008	368,780
Model B + answer score	0.979	[0.968, 0.990]	-0.022***	4,029,008	368,780
Model B + full quality controls	0.950	[0.939, 0.960]	-0.052***	4,029,008	368,780

Notes: Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

Table G.17 revisits the discrete response-time bins after adding the same three answer-quality controls used in the pooled speed specification (acceptance, first-answer score, and length). Because each row already conditions on a response-time window, the quality-controlled fits use Model A covariates plus those three signals rather than continuous log-response-time interactions. Comparing columns shows how much of the bin-wise arrival pattern survives once observable quality is held fixed.

Table G.17: Treatment Effect by Response-Time Bin, With and Without Answer-Quality Controls

Response time	Baseline HR	95% CI	Quality HR	95% CI	N
0-15 min	0.998	[0.989, 1.007]	0.953***	[0.947, 0.959]	17,412,740
15-30 min	1.004	[0.989, 1.019]	0.951***	[0.941, 0.962]	13,934,456
30-60 min	1.190***	[1.174, 1.206]	1.029***	[1.018, 1.040]	13,549,736
1-2 hr	1.056***	[1.040, 1.072]	0.965***	[0.953, 0.976]	13,211,144
2-4 hr	1.030***	[1.013, 1.047]	0.961***	[0.948, 0.974]	12,958,927
4-8 hr	1.017†	[0.999, 1.035]	0.968***	[0.953, 0.982]	12,762,008
8-12 hr	1.019	[0.995, 1.044]	0.971**	[0.951, 0.991]	12,464,162
12-24 hr	1.048***	[1.028, 1.067]	1.008	[0.993, 1.024]	12,645,932
1-3 days	1.088***	[1.069, 1.108]	1.089***	[1.071, 1.106]	12,592,667
>3 days	1.015	[0.992, 1.038]	1.041***	[1.020, 1.063]	12,391,574

Notes: Each row fits Model A to treated questions in that response-time bin plus the full no-answer control pool. Bins are cut on the actual question-to-answer latency ($T_A - T_Q$); because the observation window closes two days after the answer, every treated question retains a full two-day post-answer phase, so β_4 is identified in all bins including >3 days. HR is the answer-arrival increment (β_4). Quality columns add acceptance, first-answer score, and length.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Table G.18 probes whether the answer-quality result depends on how the vote-score control is coded. The specifications in Table G.15 enter the score linearly (winsorized at the 5th and 95th percentiles and standardized); here we re-estimate the score-only and full-quality specifications, pooled and in the newcomer stratum, under two alternatives — a signed-log (asinh) coding and ordinal bins (1–2, 3–9, and ≥ 10 votes, reference ≤ 0) — at full data. We use asinh(score) rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative (downvoted) scores, for which $\log(1 + \text{score})$ is undefined. These negative-scored cases are treated questions whose earliest reply within the window was a downvoted answer that arrived before the later non-negative answer qualifying the question as treated; because *firstAnswerScore* and the response-time clock T_{answer} are keyed to that first reply rather than to the qualifying one, both can reflect the downvoted answer rather than the one that earns treatment status. The below-one full-specification result is robust to the coding: the pooled arrival increment is 0.98 under all three codings and the newcomer counterpart 0.95, and if anything the better-specified signed-log and binned codings pull the estimate marginally *further* below one rather than back toward it. The one estimate that does not

survive re-coding is the intermediate score-only step, which is significantly above one under the linear coding (1.01) but statistically indistinguishable from one under the signed-log and binned codings; we therefore do not read it as evidence of an above-one effect.

Table G.18: Answer-Score Coding Sensitivity

Specification	Score coding	Arrival HR	95% CI	Waiting β_2	N	Events
<i>Pooled</i>						
Speed baseline	—	1.038***	[1.034, 1.042]	+0.014***	24,349,760	3,435,973
+ Vote score	Linear	1.010***	[1.006, 1.015]	-0.015***	24,349,760	3,435,973
	Asinh	1.001	[0.997, 1.005]	-0.026***	24,349,760	3,435,973
	Ordinal bins	1.003	[0.999, 1.007]	-0.023***	24,349,760	3,435,973
	Linear	0.982***	[0.978, 0.986]	-0.046***	24,349,760	3,435,973
+ Full quality controls	Asinh	0.977***	[0.973, 0.982]	-0.052***	24,349,760	3,435,973
	Ordinal bins	0.979***	[0.975, 0.983]	-0.050***	24,349,760	3,435,973
<i>Newcomer (< 1 Week)</i>						
Speed baseline	—	1.010†	[0.999, 1.021]	+0.008	4,029,008	368,780
+ Vote score	Linear	0.979***	[0.968, 0.990]	-0.022***	4,029,008	368,780
	Asinh	0.974***	[0.963, 0.985]	-0.028***	4,029,008	368,780
	Ordinal bins	0.976***	[0.965, 0.987]	-0.026***	4,029,008	368,780
	Linear	0.950***	[0.939, 0.960]	-0.052***	4,029,008	368,780
+ Full quality controls	Asinh	0.947***	[0.936, 0.957]	-0.056***	4,029,008	368,780
	Ordinal bins	0.949***	[0.938, 0.959]	-0.054***	4,029,008	368,780

Notes: Codings of the answer’s vote score, all re-expressing the same control: *Linear* is the control tabled in Table G.15 (raw *Score* winsorized at the 5th/95th percentiles and standardized); *Asinh* is the inverse hyperbolic sine $\text{asinh}(\text{score}) = \ln(\text{score} + \sqrt{\text{score}^2 + 1})$ (a signed log, concave in the score and defined for the negative scores present), likewise winsorized and standardized; *Ordinal bins* are indicators for 1–2, 3–9, and ≥ 10 votes (reference score ≤ 0). Arrival-HR stars test H_0 : HR = 1. The bin reference is score ≤ 0 and the concave coding uses asinh rather than $\log(1 + \text{score})$ because roughly two million answers carry net-negative (downvoted) scores, for which $\log(1 + \text{score})$ is undefined.

Estimand (β_4/β_2), standard-error, and Events conventions follow Table 7.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; † $p < 0.1$

Appendix H. Implementation Details

All models are estimated using the `lifelines CoxTimeVaryingFitter` with a small ridge (L2) penalizer applied for numerical stability. The penalizer strength is $\lambda = 5 \times 10^{-3}$ for main-effect models and $\lambda = 10^{-2}$ for models that include response-time interaction terms (where the interaction covariates are correlated with the base treatment indicator). These values are sufficiently small that they introduce negligible shrinkage relative to the scale of the estimated effects, but they ensure stable Hessian inversion in large risk

sets. All covariates are mean-centered before fitting; response-time interaction terms are additionally clipped to their 5th–95th percentile range and z-scored as described above. Before each fit, interval boundaries are rounded to a one-hour grid (interval start and stop times to the nearest hour) and zero-length intervals produced by the rounding are dropped. Multiple helping events by the same user within the same hour of a window therefore collapse to a single counted event. This is the “time rounding” referenced in the table notes; all reported *Events* counts are computed after it. It is also why two families of event counts differ by construction. The raw mean of 0.18 help events per window (Table 4) corresponds to 0.1411 counted events per row in the estimation sample (3,435,973/24,349,760), and the composite-outcome Events (Table 6) are smaller than the sum of their component channels, because same-hour events across channels likewise merge. Cox-table standard errors are model-based (inverse-Hessian) and are *not* cluster-robust; dependence from matched pairs, reused controls, and repeated intervals can make those intervals too narrow. For the Model A DiD we therefore also report matched-pair bootstrap 95% CIs (resampling whole `match_id` pairs) in Table F.12 — 100 replicates for the pooled effect and 500 for the newcomer (< 1 week) stratum that carries the headline result. For the pooled models (H1, spanning all experience levels), the reported point estimates are fit on the full interval frame (the fit cap is raised for the point-estimate fits), so no subsampling is applied; the default cap of approximately 8 million matched-pair user-interval rows (preserving pair integrity) applies only to the pooled bootstrap replicates, whereas the smaller newcomer stratum is fit in full within each replicate. A single 8 million-row subsample proved seed-fragile for the pooled arrival increment, which is why the reported estimate comes from a full-data fit. Table *Events* counts refer to the full analysis sample throughout, and the estimation-sample sizes are tabulated in Table E.11. Tenure-bucket-specific models (H2, H3) use all available intervals within each stratum.